Innovation Without Democratization: The Political Limits of Technological Change

Tim O'Brien

August 2026

A paper submitted in partial fulfillment of the requirements
for the degree of Master of Arts
in the Committee on International Relations

Faculty Advisor: Dr. Scott Gehlbach
Preceptor: Dr. Kara Ann Hooser

Abstract

Innovation has been positively linked to a variety of economic outcomes, including long-run development, productivity, and human capital. But its political effects have received far less attention. To address this imbalance, I investigate the effect of innovation on democracy and two intermediate outcomes, state capacity and capital mobility, using a shift-share instrumental variables strategy on a panel of 187 countries from 1995 to 2015. This design exploits quasi-random variation in global patenting growth across different fields of knowledge to identify the effects of innovation. I find that innovation exerts no detectable effect on democracy, but strongly and positively increases both state capacity and capital mobility. A one-standard-deviation increase in log patents per million residents raises state capacity by approximately half a standard deviation and capital mobility by roughly a quarter of a standard deviation. I argue that the null effect of innovation on democracy stems from the offsetting influences of these two forces. State capacity reduces the elites' costs of repression and therefore inhibits democratization, while capital mobility decreases the costs of conceding to democracy and thus promotes it. Consistent with this account, I find that innovation widens the gap between the state's capacity and society's ability to hold it accountable. A widening gap should erode democracy on its own, so paired with the null it suggests that capital mobility offsets the state's growing power. My estimates are robust across several specifications, measures, and samples.

Tim O'Brien
Committee on International Relations
University of Chicago
Chicago, IL 60637
obrienwt@uchicago.edu

Acknowledgements

This thesis would not have been made possible without the feedback and support of many people. I first want to express my deepest gratitude to my faculty advisor, Dr. Scott Gehlbach, for his unwavering patience, technical and conceptual guidance, and understanding. Any student would be lucky to have him as an advisor. I'd also like to thank my preceptor, Dr. Kara Ann Hooser, whose feedback was invaluable in shaping the structure and narrative of my paper. Her encouragement and personal support helped me through the more difficult stages of the project. I am also grateful to Nadav Kunievsky and Knowledge Lab for providing data. Finally, I want to thank my family for their love and confidence in me throughout this process. I extend particular gratitude to my father, who discussed the ins and outs of this project with me for too many hours to count.

I acknowledge the use of Claude Code and Codex to debug code and assist with the computational implementation of my analysis. This paper also benefitted from a review generated by refine.ink. All remaining work and errors are my own.

Contents

1 Introduction

Does innovation bring democracy? This question sharpens an older and more familiar discussion. Since at least Lipset (1959), the link between development and democracy has been studied again and again, and yet the literature continues to produce mixed results. Some report positive associations (e.g. Barro, 1999; Treisman, 2015), while another camp argues that both development and regime type are artifacts of specific historical conditions (e.g. Przeworski et al., 2000; Acemoglu et al., 2008). Why the divergence? One possible reason is that, just as countries differ in their *level* of development, they also differ in the *origins* of their growth. Different sources of growth may, in turn, carry distinct structural consequences, which then shape the political dynamics of democratization.

Innovation has been hailed as the engine of modern economic growth. Indeed, the advent of the steam engine and the development of the textile industry helped catapult the West into the industrial age, inaugurating a new era of technological transformation and sustained economic progress. A similarly impactful boom marked the late 19th and early 20th centuries, with the shift toward mass production and advances in the electricity, steel, fuel, and transportation industries. Paul Romer (2019) frames the impact of innovation well: "Human history teaches us, however, that economic growth springs from better recipes, not just from more cooking."

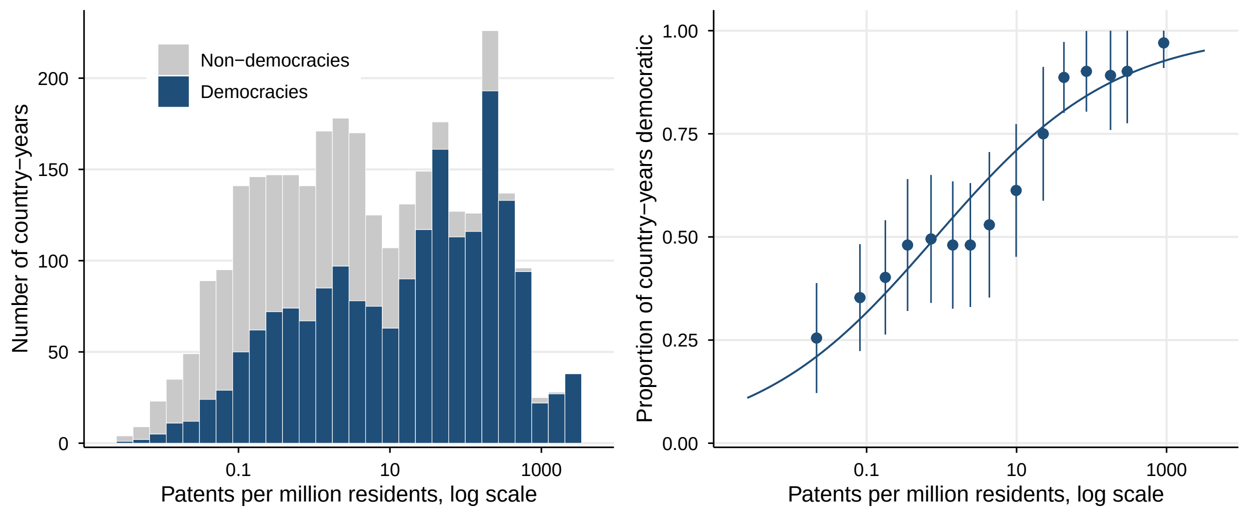
But does innovation democratize? Several technologies, such as the printing press, telegraph, and internet, have increased the scope and speed at which new information spreads, thereby lowering the costs of coordination and mobilization against autocratic rule (Manacorda and Tesei, 2020). During the Arab Spring, mobile phones and social media helped protesters overcome collective-action problems and topple autocratic incumbents in both Egypt and Tunisia. Yet only Tunisia democratized, while Egypt saw the entrenched institutions of Mubarak's regime rebuild authoritarian rule. Innovative technologies, then, might help citizens threaten autocrats, but they do not by themselves determine whether democratization will follow.

These examples concern particular technologies, but innovation is more than just a set of tools. Technological innovation is the cumulative process through which new ideas develop and build upon existing knowledge, before being transformed into usable products, methods, and systems (Schumpeter, 1934, 1942). Growth driven by this process might be quite different from growth stemming from other sources. Indeed, as Mankiw, Romer, and Weil (1992) discuss, technology is only one of many inputs to growth. Two countries at the same income level may rely on entirely different factors. For example, consider Saudi Arabia and South Korea. Despite their relatively comparable income levels, the two differ by a factor of more than 250 with regard to patents per million residents. The former relies on oil, while the latter is built on research and technology. And, as the "resource curse" literature would imply, Saudi Arabia is a full-fledged autocracy, while South Korea, by contrast, is a democracy.

Of course, two countries can differ for countless reasons, but the comparison nonetheless points to an asymmetry. A large literature explains why resource dependence, as a source of growth, tends to entrench authoritarianism (Karl, 1997; Ross, 2001), but comparatively little work investigates whether innovation-led growth promotes, hinders, or merely coincides with democracy. One possible reason for the asymmetry is that there exist compelling explanations in support of all three possibilities. The sign of innovation's effect is thus *prima facie* ambiguous.

Descriptively, innovation and democracy are strongly associated. Figure 1 shows the share of country-years that are democratic across the distribution of patents per million residents. The share rises steeply with innovation, suggesting that the modernization theory intuition holds descriptive weight. Whether this relationship is more than pure association represents the object of my study.

This paper asks whether innovation has a causal impact on democracy. Answering this question is far from straightforward. Innovative countries differ systematically from others along a variety of dimensions that also shape democracy. Moreover, democracy itself may



(A) (B)

Fig. 1. Democracy across the distribution of innovation (1995–2015)

Notes: This figure plots descriptive patterns of democracy across the distribution of patents per million residents. The unit of analysis is the country-year. Panel (A) plots the distribution of patents on a log scale, with each bar split into democratic and non-democratic observations using the Boix, Miller, and Rosato (2013) classification. Panel (B) plots the proportion of country-years that are democratic, within fifteen equally sized bins split by quantiles. The data are fit by a logistic regression with the logit link function. The whiskers represent 95 percent confidence intervals, with robust standard errors clustered at the country level. Data on patents are from PATSTAT, and data on population are from the World Bank (see Appendix A1 for details).

influence innovation, or the two could influence one another simultaneously. For these reasons, naive cross-national comparisons are unlikely to reveal a causal effect. To isolate the effect of innovation on democracy, I utilize a shift-share instrumental variables (IV) strategy in the vein of Acemoglu et al. (2016), Berkes, Manysheva, and Mestieri (2022), and Berkes and Gaetani (2023). This design exploits quasi-random variation in the worldwide expansion of various technological "fields of knowledge", to which countries are differentially exposed through their pre-sample patent citation linkages. Because these shifts are driven by worldwide growth, they should be plausibly exogenous to domestic confounders in any single country.

Armed with this strategy, I find that innovation exerts no significant effect on democracy. Further analysis suggests that this null effect is driven by innovation's tendency to increase both state capacity and capital mobility, which exert theoretically opposite effects on democracy. Guided by the models of Boix (2003) and Acemoglu and Robinson (2001, 2006), I argue that capital mobility—the *de facto* ability and motivation to move wealth outside domestic borders—lowers the elites' costs of concession to democracy, while state capacity lowers the costs of repression. On net, the two forces offset, leaving the regime type indeterminate. Consistent with this interpretation, I find that innovation widens the gap between the state's capacity and its accountability to citizens, even as rising capital mobility pushes the regime toward democracy.

This study is, to my knowledge, the first to examine the effect of innovation on democracy using modern econometric techniques. The paper contributes to several strands of literature. First, it speaks to the long-standing debate surrounding modernization theory. The null effect of innovation on democracy complements Acemoglu et al. (2008)'s broader finding that economic development has no significant influence on democracy, though I focus on innovation directly rather than growth generally. Second, it extends work on the link between the structure of economic growth and regime type. Although the resource curse literature has established a firm association between resource dependence and autocracy, other sources of growth have not been systematically investigated. My work studies one such unexplored source. Third, it complements the growing microeconomic literature on the political impacts of artificial intelligence (AI), extending its questions from individual case studies to a broad cross-country sample. Finally, it adds to the small literature directly examining the interplay of innovation and democracy. Existing studies focus narrowly on democracy's effect on innovation, with little attention paid to the opposite direction.

This question has taken on a new urgency with the rise and diffusion of AI in the 2020s. Much of the present debate over AI surrounds normative questions of mass surveillance, predictive policing, censorship, disinformation, and digital authoritarianism. Given the relative recency of this phenomenon, widespread focus has rested primarily on speculation and case studies, most of them regarding China (Xu, 2021). This paper investigates whether the patterns observed in individual countries can be generalized cross-nationally. I conclude that they do, at least in part. My study finds that earlier technologies strengthened states more than the societies they govern, even as elite wealth grows more mobile.

The remainder of the paper proceeds as follows. Section 2 outlines the theoretical framework that guides my empirical analysis. Section 3 discusses the data sources, the variables and their measurement, and the machine-learning approach used to construct fields of knowledge. Section 4 details my empirical design, including the estimation and inference procedures, specifications, and identification strategy. Section 5 reports and analyzes the results. Section 6 concludes with a summary of the core takeaways and avenues for future research.

2 Theoretical Framework

A large theoretical literature investigates the channels through which democracy emerges and persists within a state. Because innovation is at its root an economic and technological force, I organize its political effects using the redistribution-based models of regime change posited by Acemoglu and Robinson (2001, 2006) and Boix (2003). Both depict the choice of political regime as a distributive conflict between economic classes. My interest in these models here is organizational rather than technical, and thus I do not adjudicate between them. Instead, I take from them a common set of factors that may influence the emergence of democracy, and then develop how innovation could affect those factors.

In their models, two economic classes, the poor and the elite, are in contest over the power to redistribute resources. The poor favor democracy because majority rule enables greater taxation of the elite and a more equitable distribution of wealth. The elite, by contrast, want to keep taxation low and thus resist the redistribution that democracy would bring. Facing a threat of revolution from below, the elite would prefer to defuse this pressure with promises of redistribution rather than surrender power altogether. But such commitments are not credible, as the elite retain the *de jure* power to renege on their promises once the threat subsides. Anticipating this, the poor demand not just redistribution, but democracy, which would vest institutional power in the majority and ensure that the elite follow through on their commitments. Democracy is, of course, a costly concession for the elite, as it entails a loss of both wealth and future control over policy. Moreover, the elite cannot trust that the poor will bind themselves to a set level of redistribution, creating further incentive to resist democratization. Thus, the two classes face a classic commitment problem, and the elite must weigh the costs of concession to democracy against the costs of repression.

The elite's evaluation of these two costs depends on three structural factors, two of which govern the costs of concession and one that pertains to the costs of repression. The first, which Boix (2003) and Acemoglu and Robinson (2001, 2006) both emphasize, is the distribution of income. For Boix (2003), inequality has a monotonic negative effect on democratization, since highly concentrated wealth increases the expected losses under democracy. Acemoglu and Robinson (2001, 2006) argue instead that inequality's effect on democracy has an inverted U shape, where democracy is least likely at both very low and very high levels of inequality. Their logic is that the poor have little to gain from redistribution in fairly equal regimes, and the elite would rather repress than concede in highly unequal regimes. Given the disagreement between these two models, I treat the effect of inequality on democracy as theoretically ambiguous in my framework.

The second factor is the mobility of elite assets, emphasized by Boix (2003). This variable determines the extent to which the elite can shield their wealth from redistribution under democracy. When assets are less specific (i.e. more mobile), the elite can diversify their holdings internationally in order to avoid domestic taxation. Note that this channel pertains to not simply the legal rights of the elite to relocate capital, but also their actual ability and motivation to do so. Ability is shaped by the nature of the assets themselves—for instance, whether they are knowledge-based or land-based. Motivation depends on the existing domestic risks—for example, taxation or expropriation—that make holding wealth abroad an attractive option. In either case, the theory predicts that more mobile assets lower the costs of concession.

The last factor is the state's capacity to repress. Both Boix (2003) and Acemoglu and Robinson (2001, 2006) treat capacity as given: a fixed parameter that shapes elite decision-making but lies outside the model. I depart from these models by instead treating state capacity as endogenously determined by innovation. When a state has the capability to tax, monitor, and coerce its citizenry cheaply, repression becomes a viable alternative to concession, and the elite has less reason to accept democracy.

Innovation may exert an effect on democracy through any of these three channels. The main object of this paper is to connect innovation to these structural determinants of democracy, and then to examine empirically whether innovation, on net, increases or decreases the probability of democratization.

The conceptualization of innovation determines which of these channels are most relevant. Following Schumpeter (1934, 1942), I treat innovation as a cumulative process in which new knowledge builds on previous work and manifests in novel products and methods, whose value lies in their intellectual content. Two features of this conception merit focus. First, innovation is cumulative—it does not appear out of nowhere. Rather, it builds on prior knowledge, a definitional feature that my empirical strategy exploits. Second, innovation is broad, encompassing not just new products but new methods of organization, administration, and coordination. This breadth is what makes innovation a plausible influence on any of these three channels. The framework is summarized in Figure 2. The bottom three links, drawn to democracy, represent the implications of the above theory, while the top three, discussed further below, represent predicted effects of innovation on the intermediary outcomes.

2.1 Innovation and Capital Mobility

Guided by the framework above, I now investigate the channels through which innovation might affect capital mobility. I argue, building on Mokyr (2005), that innovation shifts wealth toward more mobile forms of capital by developing technology focused on knowledge, expertise, and intangible assets, untied to any particular physical location.

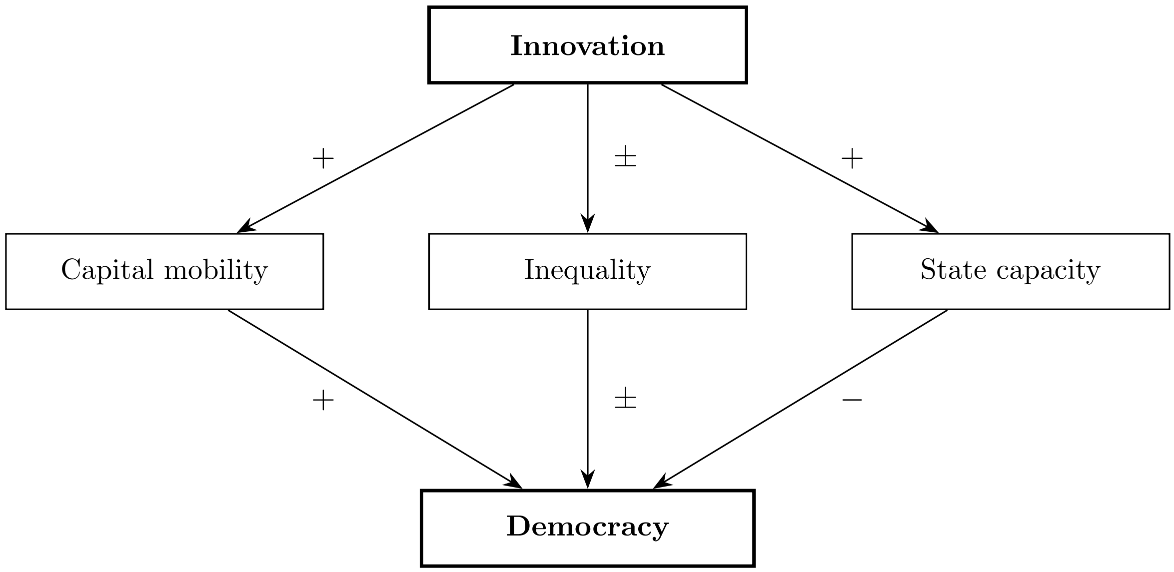


Fig. 2. Innovation and the Structural Determinants of Democracy

Notes: This figure presents a schematic representation of my basic theoretical framework. Innovation affects democracy through three distinct structural channels: capital mobility, inequality, and state capacity. Each upper link represents innovation's hypothesized effect on that particular determinant. Each lower link denotes the theoretical expected effect of that determinant on democracy. A plus sign (+) denotes a positive effect, a negative sign (−) denotes a negative effect, and a plus-or-minus sign (±) denotes an ambiguous effect.

Two well-documented empirical regularities underpin the claim that innovation increases the mobility of assets. The first is that, as economies have become more knowledge-based, the reliance on intangible assets—such as intellectual property, software, research and development, and computerized information systems—has increased in step. Corrado et al. (2013) report that intangible investment rose across all EU15 regions between 1995 and 2007, even as tangible investment stagnated or declined. This shift is not confined to advanced economies. Intangible investment has also grown rapidly in the developing world (World Intellectual Property Organization, 2025). However, the fact that capital has grown more intangible does not on its own imply that it has also become more mobile.

The second regularity makes that connection: intangible assets are more mobile than physical capital. Intuitively, intangible assets are not anchored to any given location and can therefore be moved across borders with less friction. Karkinsky and Riedel (2012) show that multinational entities file a disproportionately large share of their patents in low-tax countries, suggesting that firms do realize the advantages of mobile assets in practice.

This evidence concerns firms, but individual elites also take advantage of mobile, intangible assets to escape domestic restrictions. Alstadsæter, Johannesen, and Zucman (2019) provide evidence that offshore tax evasion is concentrated at the top economic stratum, precisely what my framework would suggest.

In all, innovation is expected to increase capital mobility by shifting wealth toward more intangible, portable forms, and indeed I find empirical support for this link. The expected effect on democracy should thus be positive. The left side of Figure 2 shows this chain schematically.

2.2 Innovation and Inequality

The second factor determining the cost of concession is the distribution of income. Here, innovation's effect is far less clear-cut. Although innovation promotes economic growth, the distributional consequences of that growth remain disputed. The "skill-biased technological change" paradigm holds that technological change raises the relative demand for skilled labor and therefore the skill premium (Katz and Murphy, 1992). The result is increased inequality.

Yet another camp contests the skill-biased thesis. Card and DiNardo (2002), for instance, show that wage inequality stabilized in the 1990s even as computerization advanced. Others argue that the effect of innovation on inequality is in any case contingent on other factors, such as the pace of educational expansion (Goldin and Katz, 2008).

Both links in the inequality channel are therefore ambiguous. Innovation may either raise or decrease inequality, and inequality has indeterminate effects on democracy. The middle links of Figure 2 represent this chain. The impact of the inequality mechanism is further diminished by innovation's positive effect on capital mobility, since mobile assets blunt the redistributive threat posed by inequality (Boix, 2003; Freeman and Quinn, 2012). For these reasons, I regard capital mobility and state capacity as the central mechanisms of my causal story, leaving inequality as an avenue for future work. In Appendix A3.9, I nonetheless report estimates probing this link, and I find no detectable effect of innovation on inequality.

2.3 Innovation and State Capacity

Recall that the cost of repression, unlike the cost of concession, is treated as exogenous by the canonical models. I instead endogenize it, arguing that this cost is chiefly determined by the state's capacity. In other words, the more capable the state, the cheaper repression becomes. I thus treat state capacity as an intermediary mechanism negatively linking innovation to democracy. Following Mann (1984), I understand capacity as the state's "infrastructural power", which includes the ability to penetrate society, monitor its citizens, and implement decisions across its territory. Two details of this power merit further note. First, state capacity is separate from accountability. A state can increase its power to tax, surveil, monitor, and coerce its citizens without growing any more accountable to them. Second, this power concerns ability, not disposition. Although heightened state capacity lowers the costs of repression, it does not necessarily force the state to use it for autocratic aims, but it does make such goals more attainable.

Innovation bears directly on state capacity. Digital surveillance, identification technology, and predictive policing algorithms lower the state's costs of monitoring and preempting citizen-level dissent, while administrative and data-processing technologies raise the efficiency with which the state can respond to threats, further lowering repression costs (Xu, 2021). Using data on facial-recognition AI in China, Beraja et al. (2023) find that local unrest increases government procurement of facial-recognition AI, and that this procurement is in turn used to stifle further discontent. This case study elucidates precisely the mechanism I posit here.

On the other hand, these same technologies may help the citizenry by lowering information costs and enhancing coordination feasibility, and these effects may increase *pressure* for accountability (Manacorda and Tesei, 2020). But the two effects are not symmetric in magnitude. Coordination and collective-action gains raise pressure for accountability only intermittently and are themselves vulnerable to manipulation by the state. Meanwhile, heightened state capacity may increase the ability of the sitting government to withstand

and live through pressure, both directly and passively.

In aggregate, I argue that innovation increases state capacity, thereby making repression cheaper and concession to democracy less necessary for the elite. The result is that the state grows more capable of coercion but no more accountable, a net negative on democracy. The right side of Figure 2 visualizes this mechanism.

2.4 The Counteracting Forces and Democratic Accountability

Of the three channels, only capital mobility and state capacity have clear theoretical links to both innovation and democracy. Innovation increases the mobility of elite assets, which lowers the cost of conceding democracy, yet it also fosters state capacity, which lowers the cost of repression. Inequality, which has ambiguous effects, does not contribute to this balance. Whether innovation democratizes or not thus depends on which force, on net, is larger.

This indeterminacy is the primary implication of my framework. From it, I draw three empirical predictions: (1) innovation increases capital mobility, (2) innovation raises state capacity, and (3) these two channels offset to produce an indeterminate effect of innovation on democracy. The third prediction warrants further explanation. A natural concern is that a null effect can stem from either offsetting underlying mechanisms or the genuine absence of any effect. My framework implies the former, and my evidence agrees. Innovation empirically raises capital mobility, so a democratizing effect is present, but innovation also widens the gap between the state's capacity and the ability of its citizens to hold it accountable, suggesting the presence of the countervailing force.

3 Data

The data under study comprise an annual unbalanced panel of 187 countries from 1995 to 2015, although some variables are missing observations across this sample.[1] The unit of analysis is the country-year.

3.1 Data Sources and Variables

As my main explanatory variable, I use the log of patents per million residents, which I construct using data on patent counts from PATSTAT and total population from the World Bank's World Development Indicators (WDI). Data for the patents per million residents measure span an unbalanced panel of 208 countries from 1995 to 2015. The panel begins in 1995 in order to leave sufficiently many pre-period years to construct my instrument's exposure shares, and it ends in 2015 because patent coverage becomes sparse. Country-years with zero recorded patents are excluded from the logged treatment. I assign patents to countries using the procedure detailed in Appendix A1.1. Patents are a standard, broadly available measure of realized inventive output, and expressing them per million residents isolates innovation intensity from a country's size. They capture formally realized output rather than innovation in its broadest sense, and thus the effects discussed below should be read as those of development driven by this particular form of innovation.

For my primary outcome variable, I use a binary measure of democracy constructed by Boix, Miller, and Rosato (2013). This measure is available for an unbalanced panel of 196 countries from 1800 to 2020. Boix, Miller, and Rosato, hereafter "BMR", code a country-year as democratic if its executive and legislature are chosen in free and fair elections and a baseline suffrage threshold is met. Although there exist several rigorous dichotomous measures of democracy, I choose the BMR measure to maximize my sample. As shown in Appendix A3.2, my results are not sensitive to alternative dichotomous and continuous

[1] The 187 countries in my analysis are those with non-missing treatment and at least one democracy outcome in each year. For each table, I provide estimates from the complete-cases sample for all variables in a given column.

measures of democracy, and the various measures are all highly correlated with one another.

I opt for a dichotomous characterization of democracy and dictatorship for reasons familiar to the political economy literature. Following Munck and Verkuilen (2002)'s framework, continuous measures of democracy tend to suffer from overburdening the concept of democracy with extraneous subconcepts, which might themselves be better studied in isolation. Often, these subconcepts either rarely have empirical referents or restrict alternative analyses. Although a minimalist conception of democracy may discard relevant details, I believe its strengths to outweigh its weaknesses.

My argument also concerns two intermediate outcomes, state capacity and capital mobility, which serve as mechanisms linking innovation to democracy. As implied by my framework, the null effect of innovation on democracy is consistent with innovation strengthening the state while leaving its accountability to citizens unchanged, even as elite wealth becomes more mobile. I therefore examine these two forces alongside democracy, together with several accountability indexes used to decompose the democratic side of that tension.

State capacity is measured primarily with the latent state capacity estimates of Hanson and Sigman (2021), which are available for 166 countries from 1960 to 2015. The Hanson and Sigman index aggregates a set of indicators spanning the state's fiscal, coercive, and administrative capacities into a single latent dimension. I check robustness by using the continuous government effectiveness index from the World Bank's Worldwide Governance Indicators (WGI) developed by Kaufmann, Kraay, and Mastruzzi (2010). This index captures perceptions regarding the quality of public services, bureaucracy, and policy implementation and formulation for 194 countries from 1996 to 2022.

Following Boix (2003), I take capital mobility to be the degree to which elites can relocate wealth outside the reach of domestic taxation. I measure it as the log of total foreign asset holdings per capita, using data from the External Wealth of Nations database. This database extends Lane and Milesi-Ferretti (2007)'s *de facto* estimates of financial integration to include 214 economies from 1970 to 2024. I prefer a *de facto* characterization of capital mobility, as opposed to a *de jure* one, because the movement of capital across borders depends on both elite *ability* and *motivation* to integrate, determined by elite choices. A statutory right to move assets that no elite exercises in practice generates no exit threat: in other words, elites who could move wealth abroad but do not would still face costs from redistribution under democracy. I also use assets because the elite diversification mechanism pertains to domestic wealth held abroad, while liabilities record foreign claims on the domestic economy.

Realized positions are, of course, an equilibrium outcome, determined by elite choices. The same holds for state capacity and democracy, which are held to be endogenous. My IV strategy, which leverages an exclusion restriction, is particularly well-suited to this setting, as it allows me to isolate the effects of innovation on these outcomes without requiring that they be exogenous. I check robustness to Chinn and Ito (2006)'s *de jure* capital-account openness index, a country's formal policy stance toward the flow of capital across its borders. Consistent with my theory, I find that innovation exerts no detectable effect on the *de jure* measure, influencing only the *de facto* indexes.

To measure democratic accountability, I use three indexes from the Varieties of Democracy (V-Dem) Institute's dataset (Coppedge et al., 2025), each available for 181 countries from 1980 to 2023. They are V-Dem's vertical, horizontal, and diagonal accountability indexes. The first measures the extent to which citizens can hold incumbents accountable, while the second measures the oversight of the executive by other institutions. The third captures the extent to which the media and civil society check the executive.

To control for factors that may jointly determine both innovation and democracy, I include several controls in alternative specifications. To avoid "bad control" problems stemming from innovation's downstream effects, I measure these covariates only in the pre-sample period, taking their means for all available observations before 1995.[2] These variables include the log of GDP per capita in constant 2015 U.S. dollars, the log of total population, trade as a share of GDP, and oil rents as a share of GDP from the World Bank's World Development

[2]For those without pre-1995 observations, I use the nearest available value through 2000.

Indicators; average years of schooling from Barro and Lee (2013); and *de jure* capital-account openness from Chinn and Ito (2006)[3]. Some specifications also include region fixed effects. The robustness of my results to other specifications is explored later in this paper.

Appendix A1.4 reports summary statistics for all variables in my analysis, as well as their accompanying sources.

3.2 Operational Classification of Fields of Knowledge

Patent data are typically organized according to classification schemes built for administrative purposes rather than analysis. The most authoritative of these, the International Patent Classification (IPC), sorts inventions according to their technical features, which do not necessarily reflect the field of knowledge to which the patent contributes. This distinction is highly relevant for my study. Because innovation develops through the synthesis of several related bodies of knowledge, any given domain of knowledge is likely to span multiple IPC subclasses. Left in their administrative form, these codes would treat parts of one knowledge domain as unrelated fields, thereby dispersing a country's exposure across many disconnected pieces of a larger, single domain. To confront this issue, I follow Berkes, Manysheva, and Mestieri (2022) (hereafter "BMM") and group the finely defined administrative categories into more general fields of knowledge.

Like BMM, I consider all available IPC classes at the four-digit level for a total of 659 classes. After dropping eight residual classes with unclear associations to specific technologies, my analysis covers 651 different IPC fields.[4] Below, I describe the clustering approach used to group these 651 IPC classes into broader fields of knowledge, which are then used in the instrument to operationalize cross-field variation.

The procedure utilizes inventor-level information to group patents into larger fields of

[3] I emphasize this measure as *de jure* in order to distinguish it from the *de facto* capital mobility measure used as an outcome. This index captures a country's formal policy stance toward the flow of capital across its borders rather than the actual volume of capital flows.

[4] The dropped subclasses are A99Z, B99Z, C99Z, D99Z, E99Z, F99Z, G99Z, and H99Z. The IPC labels these residual fields as "Subject matter not otherwise provided for in this section." As in BMM, I exclude them from my analysis because they do not pertain to clearly identified technologies.

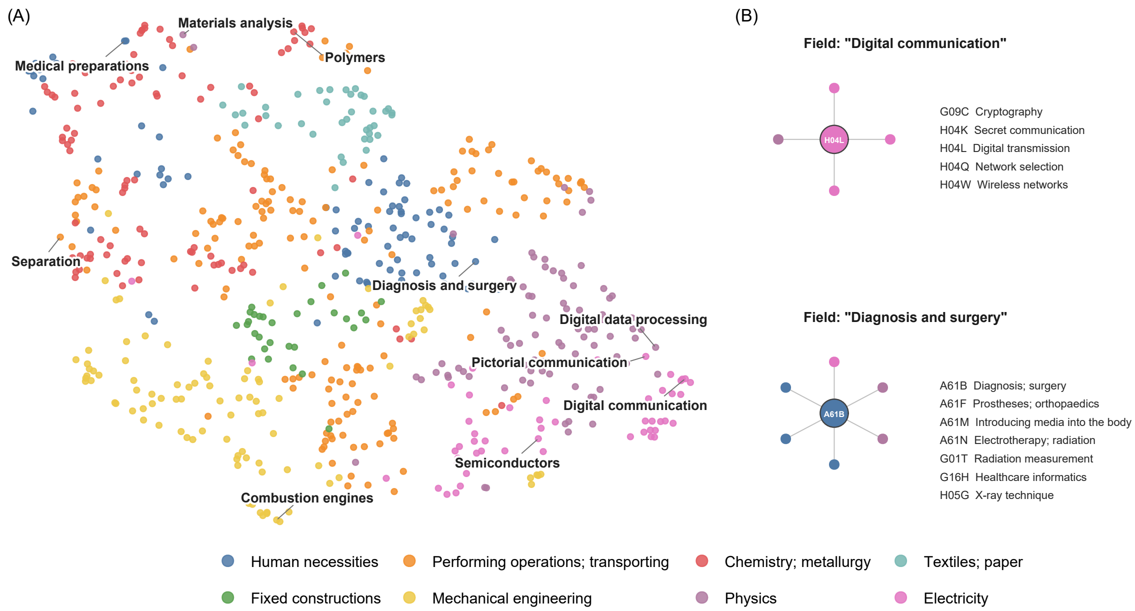
knowledge. The basic logic holds that an inventor possessing a certain corpus of knowledge is likely to produce innovations associated with multiple interrelated subclasses. As outlined above, IPC subclasses are more narrowly defined than an inventor's own domain of knowledge. To borrow an example from BMM, "a mechanical engineer specialized in brakes will most likely patent in IPCs B60T (Vehicle Brakes or Parts Thereof) and F16D (Clutches, Brakes)," and thus these two subclasses ought to be understood as components of a common underlying domain of knowledge (Berkes, Manysheva, and Mestieri, 2022, 8).

I measure the relatedness of any two subclasses by the frequency at which the same inventors patent in both. These frequencies are then converted into distances, so that subclasses sharing many inventors lie close to one another in the constructed knowledge space. I then partition the 651 IPC subclasses into distinct fields of knowledge using the *k-medoids* clustering algorithm on these distances. Appendix A1.2 provides a detailed formal discussion of this procedure.

In line with common practice, I use the silhouette method to determine the optimal number of clusters, which accounts for both within-cluster and cross-cluster distances, while also penalizing singleton clusters. The silhouette coefficient is maximized at $k = 263$.[5] The resulting output, visualized below, can be interpreted as the "knowledge space", where each cluster represents a field of knowledge.

Panel (A) in Figure 3 above shows that the IPC subclasses do tend to lie near others in the same broad IPC section, so inventor co-patenting does recover coherent technological structure. This is the structure that the clustering algorithm partitions into fields. Panel (B) shows two constructed fields as examples. Both fields include subclasses from different broad IPC sections, indicating that the constructed fields capture technological relatedness that the coarse IPC classification misses.

[5]The clustering solution has 263 fields, but one field contributes no active shock after the analysis filters used for the shift-level diagnostics. I therefore refer to the taxonomy as the $k = 263$ solution while reporting 262 active fields in the diagnostic tables. The substantive estimates use the same underlying field assignment throughout.



(A) (B)

Fig. 3. The Knowledge Space and Example Fields of Knowledge

Notes: Panel (A) is a two-dimensional UMAP embedding of the knowledge space, where each point represents one of the 651 IPC subclasses. Subclasses frequently patented by the same inventors lie close together, and each is colored by its broad IPC section. For orientation, I label some classes with the dominant field they anchor. Panel (B) shows two example fields of knowledge from the $k = 263$ clustering solution, each as its medoid subclass (center) and member subclasses.

4 Empirical Strategy

In this section, I discuss the key threats to establishing a causal effect of innovation on democracy and outline several methods of addressing them. Section 4.1 describes the estimating equations and the statistical inference procedure. I then provide a detailed account of my identification strategy and the key assumptions in Section 4.2.

4.1 Estimation

The main specification used throughout my analysis is as follows:

$$\begin{aligned} y_{it} &= \beta D_{it} + \boldsymbol{\lambda}'\bar{\mathbf{x}}_i + \delta_t + u_{it}, \\ D_{it} &= \pi Z_{it} + \boldsymbol{\gamma}'\bar{\mathbf{x}}_i + \eta_t + v_{it}. \end{aligned} \tag{1}$$

where y_{it} is the democracy status of country i in year t. The explanatory variable of interest, D_{it}, is the log of patents per million in country i at time t. $\bar{\mathbf{x}}_i$ represents a vector of observed country characteristics, fixed at their pre-sample (i.e. pre-1995) means.[6] The δ_t terms denote a full set of year fixed effects, which account for unobserved shocks common to all units in year t. Finally, u_{it} is the idiosyncratic error term, capturing all remaining time-varying unobservable determinants of democracy. I instrument for innovation using the constructed Z_{it}, as shown in the first-stage equation, which includes all the same covariates and fixed effects as the structural equation.[7]

Because each country's exposure shares are a complete decomposition of its citations across knowledge fields, the shares sum to one: $S_i = \sum_k s_{ik} = 1$ for every country. The "sum-of-shares" control that Borusyak, Hull, and Jaravel (2022) (hereafter "BHJ") require under incomplete shares is therefore unnecessary in my context, hence its omission in Equation (1).

I estimate Equation (1) by two-stage least squares, instrumenting the endogenous regressor D_{it} for reasons developed in Section 4.2. Because y_{it} is dichotomous, Equation (1) is a linear probability model (LPM), where the coefficients represent percentage-point changes in the probability of democracy. The coefficient of interest is β, which captures the estimated effect of innovation on the probability of democracy under the assumptions discussed below.

Although country fixed effects are often preferable in cross-national panel analyses, I omit them from Equation (1). A shift-share design requires that only one of the two dimensions be exogenous—either the shares or the shifts—and I take the shifts to be as-good-as-randomly assigned (Section 4.2). Under that assumption, BHJ show that unit fixed effects are not required to purge time-invariant heterogeneity. Indeed, in BHJ's own application, they include period fixed effects but never unit fixed effects (see Borusyak, Hull, and Jaravel, 2022, Table 4). Moreover, country fixed effects are costly in my application. The identifying variation is at the field level, not within countries, and a country enters only through the

[6] For the few countries with no pre-1995 observation of a given covariate, I use the nearest available value through 2000.

[7] I write Equation (1) at the country-year level for clarity's sake, but in practice I estimate the equivalent shock-level regression in order to produce valid shift-level standard errors.

predetermined exposure shares fixed at the beginning of the sample. A simple regression of the instrument on country and year fixed effects alone yields an R-squared of 0.99. As a result, including both unit and period fixed effects drives the first-stage F-statistic to nearly zero (see: Appendix A3.6). The absence of within-country identifying variation is therefore a structural feature of the setting, not an artifact of particular specification or design choices. Their omission only changes which variation identifies β, not whether the estimate is consistent.

Throughout, I report exposure-robust standard errors clustered at the field of knowledge level. Although the estimates of the linear probability model in Equation (1) are asymptotically consistent, the residuals are mechanically heteroskedastic (Angrist and Pischke, 2009). Moreover, Adão, Kolesár, and Morales (2019) show that, in a shift-share context with exogenous shifts, units with similar share-level exposure have mechanically correlated instruments. As a result, conventional standard errors over-reject the null of no association. To address both issues, I follow BHJ and conduct inference at the *shock level* rather than the unit level. This procedure residualizes y_{it} and D_{it} against the year fixed effects and controls, aggregates the residuals to the shock level using the predetermined exposure shares, and estimates the first stage and reduced form at that level. I cluster the resulting robust shock-level standard errors at the field-of-knowledge level to allow for arbitrary serial correlation within a field's shocks. In the relevant tables, I report the "effective sample size" which records the number of independent field-year observations.[8]

4.2 Identification

The main challenges in estimating the causal effect of innovation on democracy are the issues of reverse causation and unobserved confounding. Prior research suggests that democracy itself might influence innovation by protecting property rights, encouraging openness, or

[8] More precisely, the effective sample size is the inverse Herfindahl index $\left(\sum_k \bar{s}_k^2\right)^{-1}$ of the fields' average exposure shares, $\bar{s}_k = N^{-1}\sum_i s_{ik}$, which are normalized to sum to one so that each field's influence is proportional to countries' average exposure to it.

reshaping the incentives of firms and inventors. Moreover, innovative countries differ systematically from the rest, and these differences may plausibly shape democracy. Because many confounders are difficult to observe and measure, the parameter of interest cannot be identified without systematically addressing unobserved confounding.

My strategy to address innovation's endogeneity exploits variation in country-level innovation driven by forces outside that country's own political dynamics. If this variation influences the probability of democracy, the effect can be attributed to innovation rather than confounders or reverse causation. Intuitively, this design compares the democracy status of countries whose predetermined technological profiles exposed them to fast-growing global fields against those exposed to slower-growing fields. To operationalize this comparison, I construct a citation-based shift-share instrument in the vein of Acemoglu et al. (2016), Berkes, Manysheva, and Mestieri (2022), and Berkes and Gaetani (2023).

4.2.1 Instrument Construction

The instrument is constructed by interacting a country's predetermined foreign citation exposure to a given field of knowledge, the "share" component, with the log level of worldwide patenting in the same field over time, the "shift" component.

The shares operationalize a country's exposure to global innovation using PATSTAT data on domestic citations to foreign patents. As suggested by BHJ (2025), I fix this measure to the pre-sample period, before 1995, in order to preclude any possibility of global patenting shocks influencing the composition of a country's knowledge base over time, which would risk bias by correlating the instrument with contemporaneous confounders. For each country i and field k, I compute a weighted count C_{ik} of its pre-1995 citations to foreign patents in field k. Appendix A1.3 details the formulas used to construct C_{ik}. I then normalize these counts into exposure shares within each country:

$$s_{ik} \;=\; \frac{C_{ik}}{\sum_{k'} C_{ik'}}$$

so that s_{ik} is the fraction of country i's predetermined foreign-citation knowledge base attributable to field k.

Because the $k = 263$ clustering solution covers all 651 patent classes, every cited patent is assigned to a field, and the shares mechanically sum to one. Twenty-three countries—primarily microstates, poor countries, and post-communist states—have no recorded pre-1995 foreign patent citations (see Table A4 for the full list). Their shares are undefined and thus receive no weight in the shift-share estimator, although they are still included in the overall sample. Identification therefore rests on countries with a non-zero predetermined foreign-citation knowledge base.

The shift component captures the scale of global patenting in each field of knowledge over the sample period. I define it as the log of worldwide patenting for field k in year t:

$$g_{kt} \;=\; \ln\Big(\textstyle\sum_{p:t_p=t} w_{p,k}\Big)$$

where p indexes patents, t_p is the filing year of patent p, and $w_{p,k}^{k}$ is the fractional weight with which patent p is assigned to field k. The summand is thus the worldwide count of field-k patenting in year t. The shift traces the global technological frontier of field k across my sample period of 1995 to 2015.

The instrument is then the exposure-weighted sum of the shifts:

$$z_{it} \;=\; \sum_{k\in\mathcal{K}} s_{ik}\; g_{kt}.$$

The identifying variation in z_{it} is the set of common cross-field global patenting shocks g_{kt}, which affect each country in proportion to its s_{ik} values. The year fixed effects included in Equation (1) absorb the component of these shocks common to all countries in a given year. Thus, identification stems from differences across countries in their exposure to differently evolving fields.

4.2.2 The Determinants of the Global Technological Frontier

Technological progress is an uneven phenomenon. Some fields of knowledge surge over time, while others wane, and advanced economies dominate global innovation. My identification strategy exploits the differential expansion of fields, which I argue is plausibly exogenous to any single country's domestic politics. In this section, I provide descriptive evidence in support of shift exogeneity. Assumption 2 states this condition formally, and Appendix A2 reports balance tests corroborating its plausibility.

Figure 4 plots the expansion of the eight different IPC sections over the sample period. I use the IPC categorizations here for legibility, but the same dynamics are present in the constructed fields of knowledge. Panel (A) shows that, although all sections have grown over time in absolute terms, both their starting points and their rates of growth vary from section to section. Panel (B) presents the identifying variation of interest: the differential expansion across fields of knowledge, after absorbing the common upward trend with year fixed effects.

This figure elucidates several patterns. First, the technological frontier expanded on every front, indicating that the instrument's shift-level variation is not driven by a single dominant technology. This point strengthens the plausibility of the "many uncorrelated shocks" identifying assumption, which is substantiated statistically below. Second, the trajectories of different fields were markedly uneven. Sections closely tied to the information technology revolution, such as electricity and physics, saw much greater growth than relatively mature areas, such as textiles and mechanical engineering. Last, the frontier expanded as a persistent, gradual trend over decades, without sudden or country-specific breaks. This lends descriptive support to the exogeneity of the shifts, since they do not appear to respond to developments within any single country. Appendix A2 reports descriptive statistics for the shifts.

The expansion of the frontier also appears to be generated by a fairly small number of technologically advanced countries. Over the sample period, China accounts for 35 percent of worldwide patenting, followed by Japan at 26 percent and the United States at 10 percent,

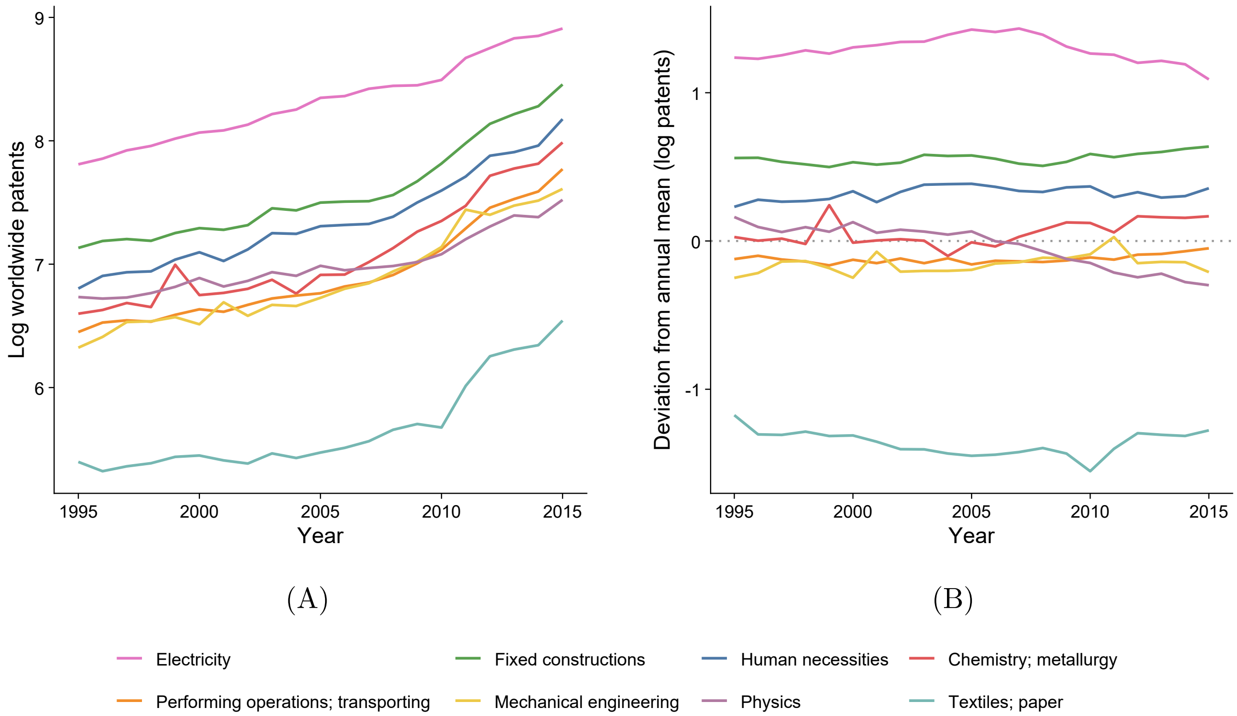


FIG. 4. THE EXPANSION OF THE GLOBAL TECHNOLOGICAL FRONTIER

Notes: This figure plots the growth of global patenting for the eight IPC sections, over the period 1995 to 2015. The instrument's identifying variation makes use of the 262 active constructed fields of knowledge, not the IPC sections. I plot the sections here for legibility purposes, although the same dynamics hold for the fields. Panel (A) plots the average log worldwide patenting of the fields in each section. Panel (B) plots the detrended variation in log patenting for each section, showing each section's deviation from the annual cross-field mean. Data are from PATSTAT, as described in Section 3.1.

while the median country accounts for under one percent.[9] The typical economy in my sample thus does not meaningfully shape the differential expansion of fields, but rather absorbs it, which suggests little correlation between a country's own innovation-related conditions and the dynamics of the global frontier.

Consistent with this conclusion, two alternative constructions of the instrument—a leave-one-out variant and a split-sample variant that rebuilds the shifts excluding the three largest producers of patents—leave the results essentially unchanged (Appendix A3). The triangulation of essentially identical results across the three instrument constructions supports that

[9]This raw share reflects the post-2008 surge in Chinese filings; as described in Section 3.2, I exclude Chinese filings when constructing the field-proximity matrix for that reason. The concentration of *filings* noted here is a separate point—it speaks only to which countries drive the global shifts, not to how fields of knowledge are defined.

the identifying variation originates on the foreign frontier.

The descriptive evidence motivates exogeneity but does not test it. Following BHJ, I test balance directly by regressing each field's shift on the exposure-weighted average of fifteen pre-1995 country characteristics. The full specification and estimates are reported in Appendix A2. The shifts are balanced on the characteristics most threatening to the democracy result: prior democracy level and democracy pre-trends. Most other measures, reported in the appendix, show similar balance.

Two sources of imbalance do appear. The first is that predetermined exposure is higher for wealthier, better educated, and more populous countries. After controlling for these pre-sample imbalances, the results remain qualitatively similar to those of the unconditional specifications.

The second calls for more scrutiny. In the decade preceding the sample, exposed countries were already trending positively in state capacity and foreign assets. At face value, these pre-trends threaten the capacity and mobility estimates, since the instrument might conflate innovation's effects with pre-existing trajectories. However, highly exposed countries were themselves also *innovating* at a rapid pace over the same decade. The instrument predicts a pre-sample rise in log patents per million of nearly the same magnitude as the rise in capacity and mobility (Appendix A2). The capacity and asset pre-trends are thus just as plausibly an early effect of innovation as they are a confound. Under the former, controlling for them would bias the estimates downwards, not up. Regardless of which reading is correct, the results I build my case on hold up after accounting for pre-trend dynamics (Appendix A2). The capacity–accountability gap is unchanged in magnitude, and the WGI measure of capacity, which begins in 1996 and therefore has no pre-sample trend, remains positive and significant. The effect on mobility, though attenuated, stays positive. Only the Hanson–Sigman capacity *level* attenuates notably, and thus I treat the gap as my primary evidence on capacity.

Altogether, these patterns bolster the case for treating the shifts as quasi-randomly assigned to units. I formalize this condition in Assumption 2 in the next subsection.

4.2.3 Identifying Assumptions

With the instrument constructed and the exogeneity of the shifts motivated, I now formalize the conditions under which 2SLS identifies the causal effects of innovation. The design inherits the standard causal logic of instrumental variables, but because identification comes from many field-level shifts rather than a single instrument, the assumptions are most naturally stated at the shock level, following BHJ. Three conditions are required for identification: instrument relevance, quasi-random assignment of shifts, and a large number of uncorrelated shocks.

Assumption 1: Relevance.

$$\mathrm{Cov}(z_{it}, D_{it}) \neq 0.$$

Assumption 1 holds that the instrument must induce variation in the endogenous regressor. This is a typical assumption for IV estimation, and its plausibility holds both intuitively and quantitatively. A country already invested in the development of a field naturally reacts more strongly to global growth in that field, so its predetermined exposure should be strongly predictive of its innovation. The first stage provides direct evidence for this condition. Across nearly all main specifications, the F-statistic exceeds conventional thresholds for instrument strength. I report a robustness check using weak-instrument-robust Anderson–Rubin inference, and my estimates remain similar to the results reported in the main text.

Assumption 2: Quasi-Random Shock Assignment.

$$\mathbb{E}[g_{kt} \mid \bar{u}_{kt}, \delta_t] = \mu_t \qquad \forall\, k,\, t,$$

Assumption 2 holds that, conditional on year fixed effects δ_t, the shifts g_{kt} are mean-independent of any unobserved exposure-weighted determinants of democracy $\bar{u}_{kt}$. In plain language, this assumption imposes that a field's global patenting growth is as-good-asrandom with respect to the political outcomes of exposed countries. Crucially, the *shares* do not need to be exogenous for identification. The identifying variation exists at the shift level, above the country.

It is helpful to couch this assumption in the more familiar exclusion restriction, which can be written as: $\mathbb{E}[u_{it} \mid z_{it}, \mathbf{x}_i, \delta_t] = 0$. This holds that, conditional on predetermined country-level characteristics and year fixed effects, a country's exposure to globally expanding technological fields affects local democracy *only* through its influence on that country's own innovation. The exclusion restriction is untestable by definition, although I argue that it is plausible for my design.

Substantively, the exclusion restriction imposes that predetermined exposure to growing global fields is orthogonal to the outcome after accounting for domestic patenting. The most plausible threat to its validity is a direct income channel, in which global growth in a country's exposed fields may raise its income through trade or knowledge spillovers, in turn affecting democracy, state capacity, or capital mobility (Berkes, Manysheva, and Mestieri, 2022). Global field growth might also reach the outcomes directly through trade demand, foreign investment, and technological imports, or through developmental and regional trends among similarly exposed countries.

Even if the exclusion restriction is violated, the reduced form is still null. Predetermined exposure to globally evolving fields does not predict democracy. Because the reduced form does not require the exclusion restriction, this null independently supports the conclusion that innovation does not affect democracy. A plausibly-exogenous bounds test in the spirit of Conley, Hansen, and Rossi (2012) likewise finds no positive democracy effect across a wide range of exclusion-restriction violations (Appendix A3.7).

Unlike a within-country dynamic panel approach (Acemoglu et al., 2019), my IV design cannot fully absorb unobserved time-invariant heterogeneity with country fixed effects. However, the exclusion restriction still mitigates heterogeneity correlated with field exposure, conditional on the shifts being as-good-as-randomly assigned.

Assumption 3: Many Uncorrelated Shocks.

$$\sum_k \bar{s}_k^2 \longrightarrow 0 \quad \text{and} \quad \mathrm{Cov}(g_{kt},\, g_{k't}) = 0 \;\; (k \neq k') \qquad \text{as } K \to \infty.$$

The last assumption concerns the number of shocks and their mutual independence. Specifically, the assumption holds that no single field dominates the exposure-weighted average of shifts. With sufficiently many shocks, the law of large numbers ensures that any by-chance correlation between the instrument and the residual vanishes in the limit. The 262 active shock-level fields, with exposure diffused across them, yield approximately 99 effective shocks, indicating that no single field dominates. I cluster at the field level to allow for arbitrary correlation among a field's shocks over time.

My focus thus far has been on the identification of innovation's effect on democracy, but the logic also extends to the two mechanisms of interest: state capacity and capital mobility. Each simply replaces democracy as the outcome in the same specification. The identifying assumptions largely carry over to these intermediate outcomes. Because excludability in a shift-share design rests on the exogeneity of the shifts, which is a property of the instrument rather than a property of any particular outcome, the evidence that shifts are as-good-as-randomly assigned supports identification of all three outcomes. I nonetheless verify excludability for each individually. An overidentification test detects no misspecification for any outcome (Appendix A3.7), and I address pre-sample trends in capacity and mobility in Appendix A2.

5 Results

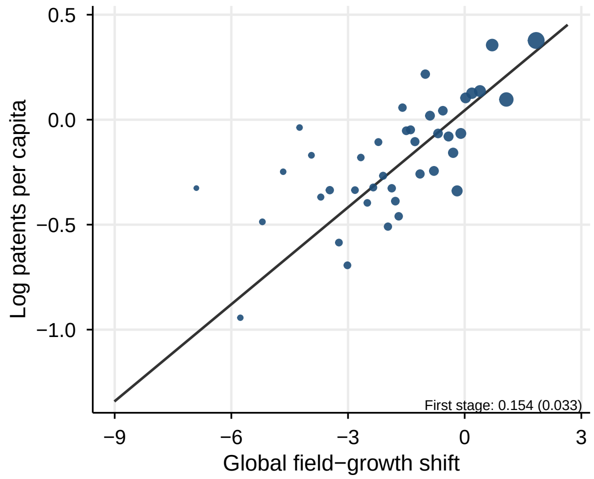
This section reports the main results of my analysis. I find that innovation exerts no detectable effect on democracy across nearly every measure, specification, and sample I examine. However, the story does not end there. Consistent with the predictions of my theoretical framework, the same design recovers large, positive, and robust effects of innovation on state capacity and capital mobility.

The remainder of this section proceeds as follows. Section 5.1 first presents the 2SLS estimates of innovation's effect on democracy, state capacity, and capital mobility. Section 5.2 then probes the robustness of my results to various possible threats to validity. Section 5.3 presents empirical evidence for innovation's tendency to widen the gap between the state's capacity and society's ability to hold it accountable, a feature I call the "capacity–accountability gap". Section 5.4 concludes with a qualitative discussion of the core takeaways of my analysis.

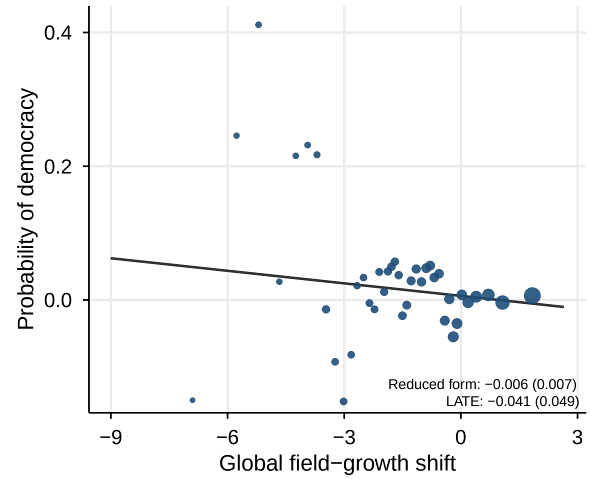
5.1 Baseline Estimates

Figure 5 provides a visual overview of innovation's effect on the three outcomes. Panel (A) shows the first stage, where log patents per million is regressed on the instrument. It demonstrates that exposure to faster-growing fields of knowledge increases domestic patenting by about 15 log points (approximately 16 percent) on average.

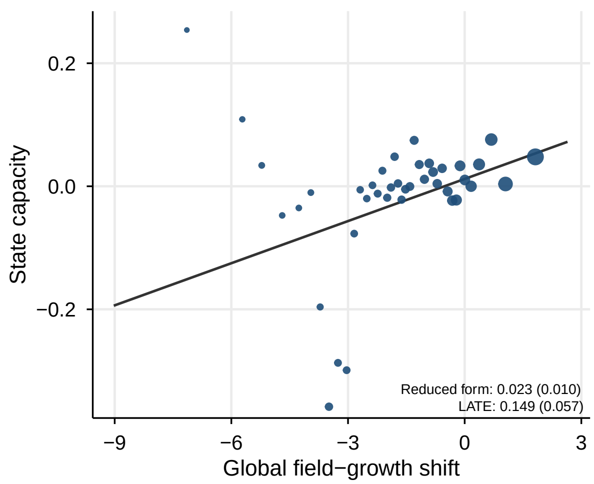
Panel (B), Panel (C), and Panel (D) display the reduced form with democracy, state capacity, and capital mobility respectively as outcomes for my preferred specification, which controls for year fixed effects, pre-sample log GDP per capita, initial democracy, and the pre-sample level of each panel's outcome (column 3 of Table 1). In Panel (B), exposure to faster-growing fields of knowledge has no significant association with democracy, with a reduced-form slope of -0.006 (standard error = 0.007). By contrast, Panel (C) shows a strong positive relationship for state capacity with a coefficient of 0.023 (standard error = 0.010), and Panel (D) shows the same for capital mobility, which has a coefficient of 0.012 (standard error = 0.006). Both coefficients are precisely estimated, significant at the five percent level. It is important to highlight the reduced form results here because they do not require the stronger exclusion restriction assumption to be informative. They imply that exposure to the expansion of the technological frontier builds the state's power and internationalizes domestic elites' wealth, while simultaneously leaving democracy relatively



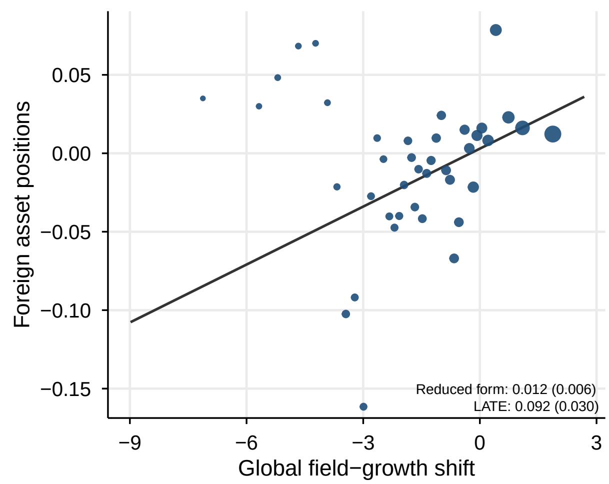
(A) First stage



(B) Reduced form: democracy



(C) Reduced form: state capacity



(D) Reduced form: capital mobility

Fig. 5. First Stage and Reduced Forms of Innovation's Effects

Notes: This figure visualizes the main results of my IV analysis. The horizontal axis is the within-year global growth of a field of knowledge. The vertical axes show the exposure-weighted outcomes—democracy, state capacity, and capital mobility—after residualizing on year fixed effects, pre-1995 log GDP per capita, the pre-1995 level of the panel's respective outcome, and pre-1995 democracy. The latter two coincide in Panel (B). Column 3 of Table 1 reports the corresponding estimates. State capacity and capital mobility are standardized. Blue points represent forty bins of field-level growth, where each is sized by its weight in the estimation. The black lines are fits to the data. Panel (A) displays the first stage, estimated on the democracy sample. First stages estimated on the samples of the other outcomes are nearly identical, so I present only one here to save space. Panels (B)–(D) plot the reduced form for each outcome and report the respective reduced-form coefficient and implied LATE in the bottom right of each panel. Exposure-robust standard errors clustered at the field level are in parentheses.

untouched. Each panel also reports the implied local average treatment effect (LATE), which I discuss during my interpretation of Table 1.

Table 1 presents formally the 2SLS estimates of Equation (1) for all three outcomes across several specifications, each controlling for different sets of plausible confounders. Note that, if the shifts are exogenous, the shares need not be, and thus these controls are not strictly necessary for identification. However, it is still possible that a country's pre-sample exposure to distinct fields of knowledge may proxy for development-related traits that also affect democracy. The controls therefore test whether the null result is sensitive to these pre-existing differences. The ladder of covariates is identical across the three panels. Common to all but one estimate, the first-stage F statistic is large and thus indicates that predetermined exposure to distinct fields of knowledge strongly predicts domestic innovative output. All models include year fixed effects, which remove within-year shocks common to all units.

Beginning with Panel (A), the coefficient on log patents per million is insignificant across every specification, indicating that innovation has no detectable effect on democracy. Column 1 includes only year fixed effects, and the estimate is negatively signed and imprecise. Column 2 controls for pre-sample log GDP per capita, the main plausible confounder, and column 3, my preferred specification, conditions on both pre-sample economic development and initial democracy to mitigate the concern that historically democratic countries accumulated systematically different knowledge bases that in turn affect contemporaneous democracy. The results remain largely similar, suggesting that these variables are not threatening confounders.

The remaining columns control for a variety of economic and demographic characteristics that may confound the relationship, including trade and *de jure* capital-account openness, oil rents, log population, and region fixed effects. The coefficient on innovation is insignificant throughout, though the region fixed effects in column 8 weaken the first stage by absorbing identifying variation.

Moving to Panel (B), I provide estimates of innovation's effect on state capacity across the same general specifications as those outlined above, substituting in the Hanson and Sigman (2021) measure as the outcome in Equation (1).

Table 1: IV Estimates of Innovation's Political and Economic Effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Democracy								
Log patents per million	−0.016	−0.065	−0.029	−0.015	−0.014	0.015	0.046	0.048
	(0.046)	(0.058)	(0.045)	(0.061)	(0.061)	(0.083)	(0.135)	(0.135)
Mean of outcome	0.626	0.629	0.627	0.629	0.627	0.664	0.664	0.664
Observations	3,055	3,007	2,997	2,609	2,601	2,291	2,291	2,291
Countries	187	184	183	151	149	126	126	126
First-stage F	10.8	18.2	23.8	18.9	29.2	27.3	13.4	9.2
Panel B. State capacity								
Log patents per million	0.229***	0.170***	0.158***	0.157**	0.158**	0.166*	0.206	0.165
	(0.049)	(0.057)	(0.060)	(0.072)	(0.071)	(0.094)	(0.141)	(0.141)
Mean of outcome	0.000	0.006	0.006	0.054	0.054	0.144	0.144	0.144
Observations	2,715	2,688	2,668	2,489	2,489	2,227	2,227	2,227
Countries	159	157	155	141	141	122	122	122
First-stage F	15.4	21.5	28.5	29.3	48.3	48.9	23.4	20.3
Panel C. Foreign asset positions								
Log patents per million	0.161**	0.026	0.081***	0.069**	0.069*	0.092**	0.086	0.069
	(0.075)	(0.050)	(0.031)	(0.034)	(0.035)	(0.045)	(0.073)	(0.085)
Mean of outcome	0.000	-0.071	-0.120	-0.158	-0.158	-0.129	-0.129	-0.129
Observations	3,175	3,049	2,910	2,576	2,575	2,267	2,267	2,267
Countries	199	190	177	149	148	125	125	125
First-stage F	10.8	23.1	28.8	22.5	28.7	29.8	13.0	10.6
Log GDP per capita		✓	✓	✓	✓	✓	✓	✓
Initial outcome level			✓	✓	✓	✓	✓	✓
Initial democracy			✓	✓	✓	✓	✓	✓
Trade and *de jure* capital openness				✓	✓	✓	✓	✓
Oil rents					✓	✓	✓	✓
Years of schooling						✓	✓	✓
Log population							✓	✓
Region fixed effects								✓
Year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Effective number of shifts	66	66	66	66	66	66	66	66

Notes: This table presents IV estimates of innovation's effect on democracy, state capacity, and capital mobility. Exposure-robust standard errors, clustered at the field level, are reported in parentheses below the coefficients on log patents per million. In all specifications, I control for year fixed effects. See the text for full descriptions of each specification. Panel (A) displays the 2SLS estimates for innovation's effect on democracy. Panels (B) and (C) report analogous estimates of the effect of innovation on state capacity and capital mobility respectively. Democracy is measured using the Boix, Miller, and Rosato (2013) index. State capacity is measured using the latent index of Hanson and Sigman (2021). Capital mobility is measured as the log of total foreign assets per capita, using data from the External Wealth of Nations. I standardize the state capacity and capital mobility measures for interpretation purposes. All controls are fixed at the pre-sample period. Control coefficients are reported in the full specification tables in Appendix A3. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Across all specifications, log patents per million is positive, and, in all but the final three columns, it remains significant at the five percent level or lower. A one log point increase in patents per million raises state capacity by 0.16 to 0.23 standard deviations, depending on the column. In my preferred specification, column 3, the coefficient on innovation is a precisely estimated 0.158 (standard error = 0.060), significant at the one percent level. The estimates remain stable as the ladder adds controls for economic openness and resource concentration in column 4 and column 5. In the remaining columns, the coefficient is the expected sign and magnitude, but becomes imprecise due to the controls and region fixed effects absorbing identifying variation.

For a sense of the magnitude, a one-standard-deviation increase in log patents per million (about three log points) raises state capacity by about half a standard deviation in my preferred specification (column 3), a sizable and precisely estimated coefficient. Because the Hanson–Sigman measure is a latent index that aggregates several fiscal, administrative, and coercive indicators into a single scale, the effect is more easily understood as a broad strengthening of the state's power rather than a change along any particular dimension. To give a tangible benchmark, a half standard deviation difference in state capacity is roughly the gap between Italy and Germany over the sample period (1.53 versus 1.99).

One caveat tempers these estimates. The Hanson–Sigman state capacity level is the outcome most sensitive to the pre-sample capacity trend documented in Section 4.2.2. However, the capacity–accountability gap, discussed in detail below, and the WGI government-effectiveness measure are unaffected. I therefore treat the gap as my primary evidence and read the level estimates here as corroborative.

Last, Panel (C) displays analogous estimates for the effect of innovation on capital mobility. Here, the results require more nuance to interpret. Unconditionally, column 1 shows that innovation has a strong, positive relationship with capital mobility (coefficient = 0.161, standard error = 0.075). In other words, an increase in log patents per million is associated with an increase in total foreign asset holdings per capita, signifying that innovation makes domestic assets more mobile across borders.

However, wealthy countries are both more innovative and hold far larger foreign asset positions. Thus, the estimate in column 1 may be confounded by prior development. Column 2 addresses this by controlling for pre-sample economic development, and the estimate becomes severely attenuated and insignificant.

Column 3 resolves the puzzle. The pattern in column 2 likely stems from capital mobility's mean-reverting dynamics. The sample period, which spans 1995 to 2015, coincides with the large-scale wave of financial globalization that marked the turn of the 21st century. With this wave, previously nonintegrated countries quickly caught up to their more integrated peers, and those that entered the sample already integrated had fairly little room to climb further. Indeed, highly exposed countries typically entered the sample highly integrated; as a result, exposure raised their capital mobility and mean-reversion attenuated it, resulting in a rough cancellation. Column 3 addresses this issue by conditioning on pre-sample levels of capital mobility. The coefficient then grows to 0.081 and is significant at the one percent level (standard error = 0.031).

After conditioning on starting position, innovation raises foreign asset positions by roughly 0.07 to 0.09 standard deviations per log point across the remaining columns, significant at the five percent level in column 4 and column 6 and at the ten percent level in column 5. As with state capacity, the final two columns are imprecise but stable in sign and magnitude, for the reasons outlined previously. The effect also attenuates, though it keeps its sign, when the pre-sample trend in foreign assets is controlled directly (Appendix A2).

To quantify the magnitude of the effect, a one-standard-deviation increase in log patents per million raises log total foreign asset holdings per capita by roughly 0.25 standard deviations in my preferred specification. Because the outcome is measured in logs, the effect amounts to roughly 0.7 log points, or about a doubling of foreign asset holdings per person. The previous benchmark applies here as well: German residents hold approximately two times more foreign assets per capita than Italian residents.

5.2 Robustness

Thus far, the primary threats to validity are the presence of omitted variables that jointly determine a country's pre-sample stock of knowledge and its subsequent political trajectories, and the possibility that global field shifts affect the outcomes through channels besides domestic innovation, violating the exclusion restriction. In this section, I systematically probe the robustness of my results to these possible sources of bias.

Following Acemoglu et al. (2019), Table 2 reports estimates of my preferred specification with each column controlling for a distinct possible threat. Column 1 presents the baseline results, equivalent to column 3 of Table 1. Columns 2–8–each control for different unit-level covariates, while columns 9-10–check robustness to shock-level threats. For all but one specification—Panel (A), column 9—the first stage F-statistic exceeds the conventional threshold.

Perhaps the clearest threat to my estimates is that countries with greater exposure to fast-growing fields were on differential trends across the sample period. For example, countries that entered the sample as democracies may have followed systematically different political paths than those that entered as autocracies, for reasons unrelated to innovation. Likewise, initially capable states may have continued strengthening their capacity, and initially integrated economies may have kept integrating. To address this, column 2 controls for an interaction between each panel's initial outcome level and year fixed effects, allowing countries to follow differential outcome trends. The estimates remain similar with this control: the effect of innovation on democracy is still insignificant, while the effects on state capacity and capital mobility are positive and precisely estimated.

Column 3 addresses the persistence of particular legal institutions in different countries. I interact an indicator of English common-law origin from La Porta, Lopez-de-Silanes, and Shleifer (2008) with year fixed effects. This indicator assigns a value of one to countries whose legal systems evolved from English common law and zero otherwise.[10] Common-law

[10] Appendix A1.5 reports the specific country-year codings for this measure.

Table 2: Robustness of the Main Estimates

		Unit-level covariates							Shock-side checks	
	Baseline (1)	Initial outcome × year (2)	Common-law dummies × year (3)	Post-communist × year (4)	Region × year (5)	Manufacturing share (6)	Inward FDI stock (7)	Spatial lags (8)	Section × year (9)	Excluding ICT fields (10)
Panel A. Democracy										
Log patents per million	−0.029	−0.018	−0.029	−0.048	−0.055	−0.077	−0.023	−0.060	−0.036	−0.026
	(0.045)	(0.053)	(0.045)	(0.058)	(0.072)	(0.088)	(0.047)	(0.049)	(0.076)	(0.051)
Mean of outcome	0.627	0.624	0.625	0.625	0.625	0.663	0.626	0.633	0.627	0.627
Observations	2,997	2,955	2,961	2,961	2,961	2,664	2,869	2,823	2,997	2,997
Countries	183	178	180	180	180	157	173	159	183	183
First-stage F	23.8	19.6	25.1	19.5	16.7	14.6	25.3	22.0	9.8	22.7
Panel B. State capacity										
Log patents per million	0.158***	0.176**	0.153**	0.177***	0.164**	0.189**	0.174***	0.158**	0.231***	0.168***
	(0.060)	(0.086)	(0.060)	(0.065)	(0.082)	(0.092)	(0.056)	(0.073)	(0.075)	(0.065)
Mean of outcome	0.006	0.011	0.011	0.011	0.011	0.079	0.022	0.051	0.006	0.006
Observations	2,668	2,661	2,661	2,661	2,661	2,382	2,625	2,543	2,668	2,668
Countries	155	154	154	154	154	134	152	138	155	155
First-stage F	28.5	14.1	33.8	35.5	28.7	22.8	34.9	28.9	16.2	25.8
Panel C. Foreign asset positions										
Log patents per million	0.081***	0.091**	0.084***	0.113***	0.100**	0.087*	0.071**	0.112**	0.113**	0.078**
	(0.031)	(0.036)	(0.028)	(0.039)	(0.049)	(0.048)	(0.032)	(0.046)	(0.044)	(0.036)
Mean of outcome	-0.120	-0.128	-0.128	-0.128	-0.128	-0.089	-0.139	-0.099	-0.120	-0.120
Observations	2,909	2,873	2,873	2,873	2,873	2,610	2,864	2,735	2,909	2,909
Countries	176	174	174	174	174	152	173	154	176	176
First-stage F	28.7	18.7	31.1	24.9	19.9	22.0	25.2	23.0	11.7	28.7

Notes: This table presents IV estimates of innovation's effects on the three outcomes, after controlling for different plausible confounders. Each column re-estimates the preferred specification, column (3) of Table 1, with the added check labeled above the column. Column (1) is the baseline. The presentation of the table mirrors that of Table 1, with panels separating each outcome, and follows Acemoglu, Naidu, Restrepo, and Robinson (2019). Columns (2)–(8) add unit-side controls: initial-outcome group-by-year fixed effects, common-law (English legal origin) by year fixed effects, post-communist dummies by year (the Soviet-dummies check of Acemoglu et al.; ex-USSR, Warsaw Pact, ex-Yugoslavia, Albania, and Mongolia), region-by-year fixed effects, the pre-1995 manufacturing share and inward FDI stock, and inverse-distance-weighted spatial lags of the outcome, the treatment, and the instrument. Columns (9)–(10) act on the shocks: technology-section-by-year fixed effects (IPC sections) and exclusion of the nine ICT fields (IPC4 majority G06/G11/H03/H04/H01L). Each cell reports the 2SLS coefficient on log patents per million residents with the mean of the outcome, the sample, and the exposure-robust first-stage F beneath; all columns are estimated at the shock level with standard errors clustered by knowledge field, with country-level controls (including the spatial lags) residualized before aggregation (Borusyak, Hull, and Jaravel, 2022). The outcome-specific batteries (alternative measures, leave-one-out and split-sample instruments, trims, and the foreign-asset-position audit) appear in Appendix A3. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

systems historically have embedded stronger protections for property rights, independent courts, and private markets—all of which plausibly influence democracy, state capacity, and capital mobility. This control has little impact on my results.

Column 4 controls for differential trends among post-communist countries, in the spirit of Acemoglu et al. (2019)'s Soviet-dummies check. Soviet and Soviet satellite countries may have followed a common transition path after 1989 for reasons unrelated to innovation. After controlling for an interaction between a post-communist dummy and year fixed effects, the estimates remain similar to the previous columns.[11]

In column 5, I verify that my estimates are not driven by shared trends among countries in the same region. In particular, I interact a regional dummy with year fixed effects to account for unobserved time-varying heterogeneity across regions.[12] This check has little effect on my baseline findings. The first stage also is essentially unaffected, remaining strong across outcomes, which suggests that regional waves do not underpin my identifying variation.

Another threat is that industrialized economies patent more frequently and tend to cite industrial fields more heavily than their less industrialized peers. Since at least Lipset (1959), industrialization has been linked to important political trajectories, and thus it may bias my estimates through downstream effects on the outcomes. Column 6 probes this threat by controlling for pre-sample manufacturing value added as a share of GDP, which I obtain data on from the World Development Indicators. Reassuringly, the estimates remain similar to the baseline.

In column 7, I consider the possibility that global growth in a country's exposed fields might attract foreign investment, which in turn could bolster state capacity, capital mobility, and democracy. To address this, I control for the pre-sample log of FDI liabilities per capita, using data from the External Wealth of Nations. Again, the results are largely unchanged, even growing in precision for the two mechanisms, relative to the previous columns.

[11] See Appendix A1.5 for a full list of the country-years I code as post-communist.

[12] To code regions, I use the World Bank's classification, which includes the following seven regions: Europe and Central Asia, Sub-Saharan Africa, Middle East and North Africa, Latin America and Caribbean, East Asia and Pacific, South Asia, and North America.

Last among the unit-level checks, column 8 addresses threats posed by spatial dependencies among countries. Democratization events occasionally spur political dynamics in neighboring countries, as in the Arab Spring. Likewise, knowledge produced by innovation crosses borders, and nearby economies may respond similarly to global innovation shocks. To address each of these threats in step, I control for spatial lags of the outcome, treatment, and instrument. For each country, the spatial lag is calculated as the average of the given variable (outcome, treatment, or instrument) across all other countries in the same year, weighted such that nearby countries contribute more to the average. In principle, this check is conservative because the contemporaneous spatial lags will absorb genuine cross-border effects of innovation alongside any confounding. Identification remains a non-issue, though, because the instrument does not rely on geography for its construction. Consistent with the ongoing pattern, the results are essentially unchanged. Democracy is insignificant, while state capacity and capital mobility are positive, similar in magnitude to the baseline, and significant at the five percent level.

Moving to the shock-level checks, columns 9–10–control for possible selection and confounding in the shifts. For instance, the growth paths of fields could reflect broad technological waves, in which case Assumption 3 would be violated. An alternative threat is that the estimates reported henceforth were driven by a handful of politically salient technologies.

To address the former, column 9 interacts IPC sections with year fixed effects. This procedure absorbs all variation between different broad technology areas and instead identifies only from within-section variation in the movement of fields. The first stage falls by roughly half in this specification, yet the results remain very similar. The effect on state capacity increases to 0.231 (standard error = 0.075), while the effect on capital mobility is similar in magnitude, at 0.113 (standard error = 0.044). Democracy remains insignificant.

On the latter threat, column 10 estimates the baseline specification excluding the nine information and communications technology (ICT) fields from the instrument. ICTs may have uniquely salient political consequences, as they are often associated with surveillance, monitoring, and collective action. Their removal then tests whether the results are driven by ICT-specific influences. The estimates reported in column 10 suggest this is not a concern, staying nearly identical to the baseline.

In the appendix, I provide further robustness checks. In Appendix A3.2, I examine different measures of democracy, including dichotomous indexes developed by Cheibub, Gandhi, and Vreeland (2010), Acemoglu et al. (2019), and Bjørnskov and Rode (2020). To relax my dichotomous characterization of democracy and dictatorship, I also consider continuous measures constructed by V-Dem, Polity, and Freedom House. In each case, innovation remains an insignificant predictor of democracy. I conduct similar checks for the other outcomes. The state-capacity effect holds across the two headline measures, while narrower indexes show consistently signed but imprecise estimates. The capital-mobility effect holds across every relevant financial position measure I examine.

In Appendix A3.1, I examine robustness to alternative measures of innovation. In particular, I re-measure the treatment using citation-weighted, originality-weighted, and generality-weighted variants to address the concern that raw patent counts reflect solely quantity rather than quality. These variants come from Trajtenberg, Henderson, and Jaffe (1997). The patterns reported in the main text are similar under these different measurements of innovation.

Beyond measurement, I additionally probe the robustness of my estimates to different IV-related decisions. Due to recent developments in the weak-instrument literature (Lee et al., 2022), it is possible that conventional instrument strength thresholds are too low. I thus re-estimate the specifications reported in Table 1 and Table 2 with Anderson–Rubin inference to understand if my results are driven by a weak instrument (Appendix A3.5). Reassuringly, the substantive conclusions remain unchanged.

Another possibility is that the instrument strength is driven by a mechanical correlation between a country's own patenting and the constructed global patenting measure used for the shifts. A leave-one-out construction of the instrument resolves this concern, and the estimates are substantively unchanged (Appendix A3.6). Because leave-one-out instruments lack a shock-level representation (Borusyak, Hull, and Jaravel, 2022), I also implement their split-sample alternative in Appendix A3.6, rebuilding the shifts from worldwide patenting excluding the three largest producers (China, Japan, and the United States, together 71 percent of worldwide patenting) for all countries. The estimates are again unchanged, indicating that neither own-country contamination nor the concentration of the frontier drives the results.

Finally, in Appendix A3.7, I exploit the large number of fields used to construct the instrument to conduct an overidentification test. I split the 263 fields into two disjoint halves and re-construct the instrument separately on each half before estimation. Provided my identifying assumptions hold, both instruments should recover the same parameter values. And, in fact, this is what I find. For each outcome, the two estimates are statistically indistinguishable, and the Hansen J test finds no evidence of misspecification for any of the three outcomes.

5.3 The Capacity–Accountability Gap

The main effects outlined above establish that both of my framework's structural forces respond to innovation: the state grows more capable, and elite wealth becomes more mobile. And the outcome that these two forces contest, democracy, remains unmoved. This section formalizes that pattern. Capital mobility reduces elite incentive to oppose democracy, while increased state capacity makes concession to democracy less necessary. This tension thus lies between the state's power to act and society's ability to check that power—a margin I call the "capacity–accountability gap". I measure this margin as the difference between standardized state capacity and a variety of standardized accountability indexes.

Figure 6 provides a useful launching pad for this analysis. In it, I plot the coefficients and 95 percent confidence intervals for eleven outcomes, grouped according to the structural force and political outcomes they measure. The coefficients are obtained by substituting each variable into the main specification as the outcome and estimating that model by 2SLS.[13] For the economic/capacity dimension, I use the GDP per capita growth rate and the government effectiveness index from the World Bank, Hanson and Sigman (2021)'s measure of state capacity, and V-Dem's measure of fiscal capacity, which captures the extent to which the state raises revenue through its tax system. For capital mobility, I use three measures of foreign asset holdings from the External Wealth of Nations database: total assets, outward direct investment, and portfolio equity, each measured per capita and in logs. For the political/accountability dimension, I use the binary BMR measure of democracy and the horizontal, vertical, and diagonal measures of accountability from V-Dem.

The effects strongly support those of the previous section. Every capacity and mobility outcome is positive, with all but fiscal capacity and portfolio equity precisely estimated. All political outcomes are statistically indistinguishable from zero. Innovation augments the administrative and coercive powers of the state and internationalizes elite assets, while leaving the mechanisms capable of checking state power untouched.

From here, I restrict my focus to state capacity and accountability. This is because the contest of relevance is that between the state and society; capital mobility shifts the incentives for elites to democratize, which affects this balance, but it is not a direct aspect of the balance. I return to the substantive implications of capital mobility after the evidence of the gap is presented.

This visual divergence above has intuitive appeal, but it is not necessarily a formal test of how the gap itself is influenced by innovation. Observing that the capacity estimates are positive and significant while the accountability estimates are insignificant does not necessarily show that the difference between them is significant. To evaluate the contrast directly, I estimate the effect of innovation on the gap itself, which I define as the standardized difference between state capacity (Hanson–Sigman index) and each accountability measure. Each difference is specified as the outcome. This procedure recovers the standard error of

[13] As these variables are meant to represent different components of the same underlying concept, the exclusion restriction is not undermined by the estimation of several outcomes.

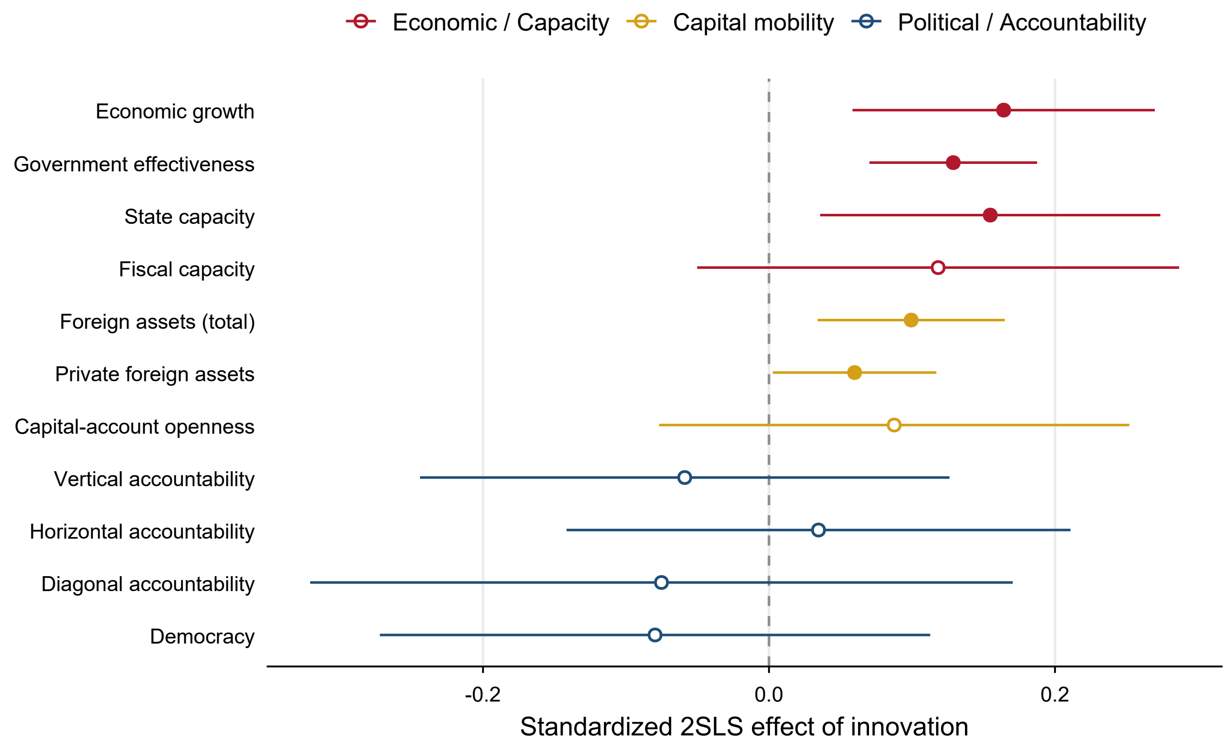


Fig. 6. Innovation's Effect on Capacity, Mobility, and Accountability

Notes: This figure visualizes the effect of innovation on various dimensions of state capacity, capital mobility, and democratic accountability, using 2SLS estimates. Red markers denote economic/capacity dimensions, gold markers capital mobility, and blue markers political/accountability outcomes. All coefficients are standardized for comparison purposes, with 95 percent confidence intervals displayed alongside. Filled-in points denote statistical significance at the 5 percent level. This figure uses exposure-robust standard errors, clustered by field.

the difference *directly* rather than by an eye test and yields a more precise estimate than a comparison of independent estimations.

The second benefit of this approach relates to identification. Because differencing the two outcomes mechanically removes any part of the instrument's influence common to both outcomes, the gap is identified under a weaker exclusion condition than either outcome individually. Suppose the exclusion restriction fails due to a development-related channel that increases both capacity and accountability. In such a case, both outcomes would be biased on their own, but the bias cancels out in the difference.[14] The remaining threat is thus

[14]This logic is commonly known as the "negative-control" mechanism, developed thoroughly by Lipsitch, Tchetgen Tchetgen, and Cohen (2010).

Table 3: Innovation and the Capacity–Accountability Gap

	Capacity	Democracy	Vert. acc.	Horiz. acc.	Diag. acc.
Panel A. Effect of Innovation on the Outcome (standardized)					
OLS	0.081***	0.038	0.022	0.047	0.021
	(0.016)	(0.032)	(0.049)	(0.031)	(0.035)
2SLS	0.152**	−0.165*	−0.044	0.017	−0.089
	(0.071)	(0.097)	(0.086)	(0.097)	(0.121)
Panel B. Effect of Innovation on the Capacity–Accountability Gap					
OLS		0.042	0.059	0.034	0.059*
		(0.030)	(0.048)	(0.029)	(0.032)
2SLS		0.317***	0.196***	0.135*	0.241**
		(0.093)	(0.070)	(0.070)	(0.101)
First-stage F	25.3	25.3	25.3	25.3	25.3
Observations	2,661	2,661	2,661	2,661	2,661
Countries	154	154	154	154	154

Notes: This table presents standardized estimates of innovation's effect on several outcomes relating to democracy, accountability, and state capacity. I use the same shift-share IV strategy as before, substituting different outcomes into Equation (1). All estimates here are from my preferred specification. Each outcome is standardized to have a mean of zero and unit variance. State capacity is measured with the Hanson and Sigman (2021) index. Democracy is measured with the BMR index. The accountability variables come from V-Dem. All columns are estimated on the same sample, the observations with all five outcomes non-missing. Panel (B) outcomes are the gaps: the capacity estimate minus each column outcome, so there is no gap for capacity itself. The standard errors are estimated directly on the gap to account for the positive correlation between each outcome and capacity. The 2SLS standard errors are exposure-robust, clustered at the field level, and the OLS standard errors are robust, clustered by country, both in parentheses. The first stage is the same for both panels. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

differential leakage, though the leakage would have to be implausibly large to change the conclusions. In Appendix A3.7, I show that the gap estimate is invariant to common direct effects across the bounded range, and it remains positive and significant under substantial differential leakage.

Table 3 reports estimates of innovation's effect on the gap. To ensure the gaps are internally consistent, all columns are estimated on a common sample.[15]

In Panel (A), state capacity rises by 0.152 standard deviations per log point, significant at the five percent level. The democracy estimate is negative and marginally significant, though one case of marginal significance in this study is to be expected with multiple testing. All

[15]Specifically, I require that all four outcomes are not missing.

accountability measures are precise zeros.

Panel (B) reports the gaps. Innovation widens the capacity–democracy gap by 0.317 standard deviations per log point (standard error 0.093), significant at the one percent level. It also widens the capacity gap with vertical, horizontal, and diagonal accountability by 0.196 (standard error = 0.070), 0.135 (standard error = 0.070), and 0.241 (standard error = 0.101) respectively. Each gap formally tests the equality of innovation's effects across outcomes, and equality is rejected for all outcomes, though only at the ten percent level for the horizontal-accountability gap. The F-statistics again indicate that the instrument is sufficiently strong.

Having established the gap, I now return to capital mobility. Unlike capacity and accountability, mobility is not a side of the state–society contest. Instead, it pertains to elites and their incentives to concede to regime change. My framework implies that increases in mobility should, holding all else constant, favor democratization. Yet democracy does not move. The gap helps explain why. On its own, a widening state–society gap should erode democracy, since the state's power would grow faster than society's ability to check it. That I find a null effect of innovation on democracy thus suggests a countervailing force, which I argue to be capital mobility. Paired with the democracy null, the widening gap is indirect evidence that mobility counterbalances state capacity.

5.4 Discussion

With the statistical results in mind, I now move to a discussion of their substantive meaning and implications. As my framework anticipates, I find a null effect of innovation on democracy and positive effects on both state capacity and capital mobility, with these two forces pulling democracy in opposite directions. The capacity–accountability gap reinforces this reading, and the large, robust effects on state capacity and capital mobility indicate that the null does not reflect a weak-instrument problem. In Appendix A3.10, I show that the null also holds across subgroups, suggesting that innovation likely does not exert heterogeneous effects, although these results are tenuous due to low statistical power.

Consider first the capacity side of the tension. By the logic of my framework, a state that grows more capable but no more accountable finds repression cheaper and concession to democracy less necessary. China became the world's largest patent filer and one of its most powerful states over the course of my sample period, and yet it became no more democratic. In many ways, it became more authoritarian, not less. The general null effect I estimate renders that case part of a broader pattern. China has used innovation to strengthen instruments of surveillance and control, including social-credit and facial-recognition systems.

Moving to the mobility side, innovation shifts elite wealth toward less specific assets, and therefore lowers the costs of concession to democracy. On its own, this process should increase the probability of democratization. The substantive implication of this mechanism is subtler than just its counterbalancing effects on democracy. As elite assets become more mobile, the state loses control over the wealth it might otherwise tax, even as its capacity to tax, monitor, and coerce the non-elites grows. Innovation thus loosens the state's grip on the elite while tightening its grip on the rest.

6 Conclusion

This study has found that innovation exerts no effect on democracy, a result robust across nearly every specification, sample, and measure I examine. This finding stands in contrast to intuition and descriptive trends. Innovation and democracy alike are often associated with notions of free expression, creativity, the creation and diffusion of knowledge, and liberalism. But my analysis suggests that any apparent effect of innovation on democracy is coincidental, an artifact of either reverse causation or joint determination by other variables. Once these features are accounted for, the effect disappears. State capacity and capital mobility, by contrast, respond strongly to innovation. I argue that the democracy null reflects an underlying balance between these forces, with mobility increasing prospects of democracy and state capacity decreasing them.

These findings carry relevant policy implications. Namely, innovation should be encouraged but carefully monitored. A more capable state is not necessarily a more autocratic one. But if a country is both capable and authoritarian, this capacity may help the regime withstand pressures for democratization (Albertus and Menaldo, 2012). A parallel caution should surround capital mobility. Innovation encourages the wealthiest to move assets abroad, narrowing the tax base and thereby weakening the state's power to redistribute. Policies promoting innovation should therefore be paired with explicit limits on state use of these technologies, in addition to measures that keep mobile wealth taxable.

My analysis could be extended in several directions. A longer panel would facilitate a better understanding of innovation's long-run effects, which might differ from the estimates reported in my two-decade-long sample. Subnational analysis is another promising avenue, as it could accommodate country fixed effects and reveal possible sectoral or local effects that cancel out on a national scale. Discretizing the measure of innovation might also enrich my findings. The theory presented in Section 2 suggests that innovation may promote democracy only when the economic benefits it brings outweigh the political losses the elites would incur under democracy (Acemoglu and Robinson, 2000, 2001). A discrete treatment that operationalizes this threshold might uncover relevant effects. Finally, a more thorough examination of possible bidirectionality between innovation and democracy would mesh well with the equilibrium-focused framework in which the two shape each other, rather than the unidirectional design I adopt here.

In all, the research agenda for innovation and democracy is bright. The rise of artificial intelligence in the 2020s has reopened the question of technology's political consequences with a new urgency and fervor. If the effects I document here hold, then perhaps AI represents another force that strengthens states, including autocracies, faster than it empowers their citizens to check government power. A greater understanding of when, why, and how technological innovation can be shaped to build democratic accountability is a relevant and important question for the next decade of political economy research.

References

Acemoglu, Daron. 2002. "Directed Technical Change." *The Review of Economic Studies* 69 (4): 781–809.

Acemoglu, Daron, David Autor, David Dorn, Gordon H. Hanson, and Brendan Price. 2016. "Import Competition and the Great US Employment Sag of the 2000s." *Journal of Labor Economics* 34 (S1): S141–S198.

Acemoglu, Daron, Simon Johnson, James A. Robinson, and Pierre Yared. 2008. "Income and Democracy." *American Economic Review* 98 (3): 808–842.

———. 2009. "Reevaluating the Modernization Hypothesis." *Journal of Monetary Economics* 56 (8): 1043–1058.

Acemoglu, Daron, Suresh Naidu, Pascual Restrepo, and James A. Robinson. 2019. "Democracy Does Cause Growth." *Journal of Political Economy* 127 (1): 47–100.

Acemoglu, Daron, and James A. Robinson. 2000. "Political Losers as a Barrier to Economic Development." *American Economic Review* 90 (2): 126–130.

———. 2001. "A Theory of Political Transitions." *American Economic Review* 91 (4): 938–963.

———. 2006. *Economic Origins of Dictatorship and Democracy.* New York: Cambridge University Press.

Adão, Rodrigo, Michal Kolesár, and Eduardo Morales. 2019. "Shift-Share Designs: Theory and Inference." *The Quarterly Journal of Economics* 134 (4): 1949–2010.

Albertus, Michael, and Victor Menaldo. 2012. "Coercive Capacity and the Prospects for Democratization." *Comparative Politics* 44 (2): 151–169.

Alstadsæter, Annette, Niels Johannesen, and Gabriel Zucman. 2019. "Tax Evasion and Inequality." *American Economic Review* 109 (6): 2073–2103.

Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion.* Princeton: Princeton University Press.

Barro, Robert J. 1999. "Determinants of Democracy." *Journal of Political Economy* 107 (S6): S158–S183.

Barro, Robert J., and Jong-Wha Lee. 2013. "A New Data Set of Educational Attainment in the World, 1950–2010." *Journal of Development Economics* 104:184–198.

Beraja, Martin, Andrew Kao, David Y. Yang, and Noam Yuchtman. 2023. "AI-tocracy." *The Quarterly Journal of Economics* 138 (3): 1349–1402.

Berkes, Enrico, and Ruben Gaetani. 2023. "Income Segregation and the Rise of the Knowledge Economy." *American Economic Journal: Applied Economics* 15 (2): 69–102.

Berkes, Enrico, Kristina Manysheva, and Martí Mestieri. 2022. *Global Innovation Spillovers and Productivity: Evidence from 100 Years of World Patent Data.* Working Paper 2022-15. Federal Reserve Bank of Chicago.

Bjørnskov, Christian, and Martin Rode. 2020. "Regime Types and Regime Change: A New Dataset on Democracy, Coups, and Political Institutions." *The Review of International Organizations* 15 (2): 531–551.

Blattman, Christopher, Scott Gehlbach, and Zeyang Yu. 2025. *The Personalist Penalty: Varieties of Autocracy and Economic Growth.* Working Paper 34093. National Bureau of Economic Research.

Boix, Carles. 2003. *Democracy and Redistribution.* New York: Cambridge University Press.

Boix, Carles, Michael Miller, and Sebastian Rosato. 2013. "A Complete Data Set of Political Regimes, 1800–2007." *Comparative Political Studies* 46 (12): 1523–1554.

Boix, Carles, and Susan C. Stokes. 2003. "Endogenous Democratization." *World Politics* 55 (4): 517–549.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2022. "Quasi-Experimental Shift-Share Research Designs." *The Review of Economic Studies* 89 (1): 181–213.

———. 2025. "A Practical Guide to Shift-Share Instruments." *Journal of Economic Perspectives* 39 (1): 181–204.

Card, David, and John E. DiNardo. 2002. "Skill-Biased Technological Change and Rising Wage Inequality: Some Problems and Puzzles." *Journal of Labor Economics* 20 (4): 733–783.

Cheibub, José Antonio, Jennifer Gandhi, and James Raymond Vreeland. 2010. "Democracy and Dictatorship Revisited." *Public Choice* 143 (1): 67–101.

Chinn, Menzie D., and Hiro Ito. 2006. "What Matters for Financial Development? Capital Controls, Institutions, and Interactions." *Journal of Development Economics* 81 (1): 163–192.

Conley, Timothy G., Christian B. Hansen, and Peter E. Rossi. 2012. "Plausibly Exogenous." *The Review of Economics and Statistics* 94 (1): 260–272.

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, et al. 2025. *V-Dem Codebook v15.* Varieties of Democracy (V-Dem) Project.

Corrado, Carol, Jonathan Haskel, Cecilia Jona-Lasinio, and Massimiliano Iommi. 2013. "Innovation and Intangible Investment in Europe, Japan, and the United States." *Oxford Review of Economic Policy* 29 (2): 261–286.

Enikolopov, Ruben, Alexey Makarin, and Maria Petrova. 2020. "Social Media and Protest Participation: Evidence from Russia." *Econometrica* 88 (4): 1479–1514.

Freedom House. 2024. *Freedom in the World.* Freedom House, Washington, DC.

Freeman, John R., and Dennis P. Quinn. 2012. "The Economic Origins of Democracy Reconsidered." *American Political Science Review* 106 (1): 58–80.

Geddes, Barbara, Joseph Wright, and Erica Frantz. 2014. "Autocratic Breakdown and Regime Transitions: A New Data Set." *Perspectives on Politics* 12 (2): 313–331.

Goldin, Claudia, and Lawrence F. Katz. 2008. *The Race between Education and Technology.* Cambridge, MA: Belknap Press of Harvard University Press.

Hanson, Jonathan K., and Rachel Sigman. 2021. "Leviathan's Latent Dimensions: Measuring State Capacity for Comparative Political Research." *The Journal of Politics* 83 (4): 1495–1510.

Karkinsky, Tobias, and Nadine Riedel. 2012. "Corporate Taxation and the Choice of Patent Location within Multinational Firms." *Journal of International Economics* 88 (1): 176–185.

Karl, Terry Lynn. 1997. *The Paradox of Plenty: Oil Booms and Petro-States.* Berkeley: University of California Press.

Katz, Lawrence F., and Kevin M. Murphy. 1992. "Changes in Relative Wages, 1963–1987: Supply and Demand Factors." *The Quarterly Journal of Economics* 107 (1): 35–78.

Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi. 2010. *The Worldwide Governance Indicators: Methodology and Analytical Issues.* Policy Research Working Paper 5430. World Bank.

La Porta, Rafael, Florencio Lopez-de-Silanes, and Andrei Shleifer. 2008. "The Economic Consequences of Legal Origins." *Journal of Economic Literature* 46 (2): 285–332.

Lane, Philip R., and Gian Maria Milesi-Ferretti. 2007. "The External Wealth of Nations Mark II: Revised and Extended Estimates of Foreign Assets and Liabilities, 1970–2004." *Journal of International Economics* 73 (2): 223–250.

Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-ratio Inference for IV." *American Economic Review* 112 (10): 3260–3290.

Lipset, Seymour Martin. 1959. "Some Social Requisites of Democracy: Economic Development and Political Legitimacy." *American Political Science Review* 53 (1): 69–105.

Lipsitch, Marc, Eric Tchetgen Tchetgen, and Ted Cohen. 2010. "Negative Controls: A Tool for Detecting Confounding and Bias in Observational Studies." *Epidemiology* 21 (3): 383–388.

Manacorda, Marco, and Andrea Tesei. 2020. "Liberation Technology: Mobile Phones and Political Mobilization in Africa." *Econometrica* 88 (2): 533–567.

Mankiw, N. Gregory, David Romer, and David N. Weil. 1992. "A Contribution to the Empirics of Economic Growth." *The Quarterly Journal of Economics* 107 (2): 407–437.

Mann, Michael. 1984. "The Autonomous Power of the State: Its Origins, Mechanisms and Results." *European Journal of Sociology* 25 (2): 185–213.

Marshall, Monty G., and Ted Robert Gurr. 2020. *Polity5: Political Regime Characteristics and Transitions, 1800–2018.* Center for Systemic Peace, Vienna, VA.

Mokyr, Joel. 2005. "Long-Term Economic Growth and the History of Technology." In *Handbook of Economic Growth,* edited by Philippe Aghion and Steven N. Durlauf, vol. 1B, 1113–1180. Amsterdam: Elsevier.

Munck, Gerardo L., and Jay Verkuilen. 2002. "Conceptualizing and Measuring Democracy: Evaluating Alternative Indices." *Comparative Political Studies* 35 (1): 5–34.

Przeworski, Adam, Michael E. Alvarez, José Antonio Cheibub, and Fernando Limongi. 2000. *Democracy and Development: Political Institutions and Well-Being in the World, 1950–1990.* New York: Cambridge University Press.

Przeworski, Adam, and Fernando Limongi. 1997. "Modernization: Theories and Facts." *World Politics* 49 (2): 155–183.

Romer, Paul M. 2019. *Economic Growth.* https://paulromer.net/economic-growth/.

Ross, Michael L. 2001. "Does Oil Hinder Democracy?" *World Politics* 53 (3): 325–361.

———. 2008. "Oil, Islam, and Women." *American Political Science Review* 102 (1): 107–123.

Schumpeter, Joseph A. 1934. *The Theory of Economic Development.* Cambridge, MA: Harvard University Press.

———. 1942. *Capitalism, Socialism and Democracy.* New York: Harper & Brothers.

Solt, Frederick. 2020. "Measuring Income Inequality across Countries and over Time: The Standardized World Income Inequality Database." *Social Science Quarterly* 101 (3): 1183–1199.

Trajtenberg, Manuel, Rebecca Henderson, and Adam Jaffe. 1997. "University versus Corporate Patents: A Window on the Basicness of Invention." *Economics of Innovation and New Technology* 5 (1): 19–50.

Treisman, Daniel. 2015. "Income, Democracy, and Leader Turnover." *American Journal of Political Science* 59 (4): 927–942.

World Intellectual Property Organization. 2025. *World Intellectual Property Report 2025.* WIPO, Geneva.

Xu, Xu. 2021. "To Repress or to Co-opt? Authoritarian Control in the Age of Digital Surveillance." *American Journal of Political Science* 65 (2): 309–325.

A1 Data and Construction

A1.1 Assignments of Patents to Countries

My procedure for assigning patents to countries is largely similar to that outlined by BMM (2022, section 2.1). I first compile information on patent filing years, countries of inventors and applicants, citation histories, and technological classes.

A patent is assigned to a country based on the inventor's country of residence. When the inventor's country is unavailable, I default to the country of the patent applicant. If neither the inventor country nor applicant country is available, the country of the filing office is used (excluding regional filing offices). Inventions with inventors or applicants from multiple countries are divided fractionally across those countries in proportion to the number of inventors or applicants. Each inventor/applicant is assumed to have contributed equally to the patent.

For instance, suppose a patent has a total of five inventors, three from the US and two from Canada. In this case, the patent would be split between the US and Canada with weights of 0.60 and 0.40. In cases where the patent was filed in multiple offices, I include only the earliest application within the patent's DOCDB family to avoid double-counting. For the citation aspect of the instrument, I do not apply this double-counting filter because a patent family's citations are typically spread across different offices.

A1.2 Construction of Fields of Knowledge

This section describes the machine-learning procedure used to group IPC4 subclasses into fields of knowledge, currently summarized in Section 3.2 of the main text.

I construct a probability matrix $T_{651\times 651}$ in accordance with the 651 IPC subclasses under study. Each element T_{ij} represents the probability that an inventor patents in a subclass i conditional on having also patented in subclass j:

$$T_{ij} = \frac{N_{ij}}{N_j},$$

where N_{ij} is the number of inventors who have patented in both i and j, and N_j is the number who have patented in j. For off-diagonal elements, where $i \neq j$, T_{ij} captures the extent to which inventors who patent in subclass j also tend to patent in subclass i.[16]

The counts N_{ij} and N_j are computed over all inventors recorded in PATSTAT, subject to two exceptions. First, I exclude patents filed at the Chinese office when constructing the matrix due to the post-2008 surge in Chinese patenting, which reflects a rise in quantity rather than quality and thus distorts the proximity matrix. The fields of knowledge are nonetheless applied to all countries, including China, in the analysis.

Second, I exclude inventors associated with more than 50 distinct IPC subclasses. An individual inventor patenting in such a large range of subclasses is implausible, and manual inspection confirms that such cases are overwhelmingly firms, institutions, or other entities rather than individuals.[17] I impose this cap because the proximity matrix measures the

[16]The diagonal elements of $T_{651\times 651}$ are set to equal one (i.e. $i = j$). Also note that the matrix is not generally symmetric because the denominator N_j may differ (and often does differ) across subclasses. Put differently, $T_{ij} \neq T_{ji}$ in general because N_j need not equal N_i. Intuitively, a narrow subclass may be most closely related to a different broader subclass, even if that broader subclass is not primarily related to the narrower one.

[17]PATSTAT identifies inventors and patent applicants using the `person_id` variable, a surrogate key assigned to each unique combination of the applicant/inventor name, original-language name, address, and country code. Thus, a `person_id` is not necessarily an individual—it may be a firm, university, or a combination of multiple individuals with the same identifying characteristics. I exclude `person_id` codes associated with more than 50 distinct IPC subclasses when constructing the proximity matrix T. Above this threshold, a large share of these `person_id` records also appear as patent applicants rather than solely as inventors. Concretely, the share of IDs that file as an applicant rises monotonically with the number of distinct fields

relatedness of patents through the individuals that work across different subclasses, and that interpretation holds only for individual inventors. Firms or institutions may patent in multiple fields for reasons unrelated to the technology itself (e.g. for organizational or commercial purposes).

In this paper, the 651 IPC subclasses are then grouped into broader fields of knowledge using the *k-medoids* clustering algorithm, consistent with BMM. Because the *k-medoids* algorithm requires a symmetric measure of dissimilarity, I first convert the proximity matrix T into a distance matrix D, such that smaller values of D_{ij} indicate greater relatedness among technologies in subclasses i and j. Specifically, for the off-diagonal-pairs, I define

$$D_{ij} = 1 - (T_{ij} + T_{ji}) = D_{ji}$$

so that $D_{ij} = D_{ji}$. The term $T_{ij} + T_{ji}$ gives a symmetric measure of proximity between i and j, and subtracting this sum from one converts the proximity to a distance. For 19 of the 211,575 (off-diagonal) subclass pairs (approximately 0.01 percent), $T_{ij} + T_{ji} > 1$, which then makes D_{ij} negative. Since a negative distance is nonsensical, I set these entries equal to zero.

The subclasses are then partitioned into distinct fields of knowledge by applying *k-medoids* clustering to D. For a given number of clusters k, the algorithm selects k IPC subclasses as medoids and assigns each remaining subclass to the nearest medoid using the distances in D. This process continues iteratively until the total within-cluster distance has been minimized.[18]

I select the number of fields k by maximizing the average silhouette coefficient of the

spanned, rising from 15 percent among IDs in five fields or fewer, to 58 percent at 31 to 50 fields, and 79 percent above fifty. Because applicants are often firms, universities, or other institutions, this pattern suggests that many such records do not correspond to individual inventors. Note that this restriction solely affects the construction of T and does not remove any patent observations from the following stages of analysis. The 50-class threshold I impose removes only 0.3 percent of all inventor records. To my knowledge, BMM do not apply an analogous restriction.

[18]There are advantages and disadvantages to all clustering methods. In my application, the *k-medoids* algorithm is a sensible choice, as it operates on actual distance metrics in D rather than on computed cluster means, as in *k-means*. The primary tradeoff is that it can only assign subclasses to precisely one cluster.

resulting partition. Singleton clusters are assigned a silhouette of zero, following BMM. The silhouette varies little across a wide range of plausible k and reaches its maximum at $k = 263$, which I choose as my baseline to construct the instrument.

A1.3 Instrument Construction

This appendix details the formal construction of the exposure shares summarized in Section 4.2.1. The shifts and combined shift-share instrument are defined in the main text.

The shares are built using PATSTAT data on pre-1995 domestic citations to foreign patents. Formally, the weighted count of pre-1995 citations from country i to foreign patents in field k is defined as:

$$C_{ik} = \sum_{\substack{(a,b):\, t_a<1995 \\ a\neq b}} w^c_{a,i} \left(\sum_{c\neq i} w^c_{b,c}\, w^k_{b,k} \right),$$

where $w^c_{a,i}$ is the share of the citing patent a belonging to country i, $w^c_{b,c}$ is the share belonging to foreign country c, and $w^k_{b,k}$ denotes the share of the cited patent b belonging to field k. The fractional component of $w^c_{a,i}$ stems from the fact that a patent can have inventors from multiple different countries, and that same logic underlies $w^c_{b,c}$. For $w^k_{b,k}$, a patent can span several fields. Note that citations between patents belonging to the same DOCDB family are excluded from the share construction, as are, naturally, cases in which the cited patent belongs to country i itself ($c = i$).

The weighted count C_{ik} captures the extent to which country i's pre-sample knowledge base drew on foreign patents in field k. This count grows larger when a country cites many foreign patents in the given field and shrinks when it does not. Normalizing the count within country gives the following exposure shares s_{ik} defined in the main text. As an example, suppose a country's pre-sample foreign citations were divided across three fields: 50 percent to computing, 30 percent to pharmaceuticals, and 20 percent to telecommunications. That country would then carry exposure shares of 0.50, 0.30, and 0.20 for those fields.

A1.4 Summary Statistics

Table A1 reports summary statistics for all variables of my analysis, organized by concept into seven panels: innovation (Panel A), democracy (Panel B), accountability and constraints (Panel C), state capacity (Panel D), foreign asset positions (Panel E), inequality (Panel F), and economic and demographic characteristics (Panel G).

Innovation data are drawn from the European Patent Office's Worldwide Patent Statistical Database (PATSTAT).

Democracy data come from Boix, Miller, and Rosato (2013) [BMR], Acemoglu et al. (2019) [ANRR], Cheibub, Gandhi, and Vreeland (2010) [CGV], Bjørnskov and Rode (2020) [BR], Geddes, Wright, and Frantz (2014) [GWF], Freedom House [FH], the Polity5 project [Polity], and the Varieties of Democracy project [V-Dem].

State-capacity data come from Hanson and Sigman (2021) [H–S], the World Bank's Worldwide Governance Indicators [WGI], and the Varieties of Democracy project [V-Dem].

Capital mobility data are the *de facto* foreign-asset positions of the External Wealth of Nations database, which extends the Lane and Milesi-Ferretti (2007) data, and the *de jure* Chinn–Ito index of capital-account openness (Chinn and Ito, 2006).

Data on inequality are sourced from the Standardized World Income Inequality Database (Solt, 2020) [SWIID].

For the predetermined controls, I use log GDP per capita (`NY.GDP.PCAP.KD`), oil rents as a share of GDP (`NY.GDP.PETR.RT.ZS`), trade as a share of GDP (`NE.TRD.GNFS.ZS`), and log population (`SP.POP.TOTL`) from the World Bank's World Development Indicators, together with average years of schooling from Barro and Lee (2013).

Table A1: Summary Statistics

	N	Mean	SD	Min	Max
Panel A: Innovation					
PATSTAT Log Patents per Million	3,319	1.79	3.08	−5.97	8.60
PATSTAT Log Citation-Weighted Patents	3,394	3.77	3.43	−5.88	11.28
PATSTAT Log Originality-Weighted Patents	2,982	0.38	3.07	−7.83	6.91
PATSTAT Log Generality-Weighted Patents	2,941	0.39	3.08	−7.74	7.09
PATSTAT Log Family-Size-Weighted Patents	3,394	2.60	3.20	−6.57	9.68
Panel B: Democracy					
BMR Democracy	18,293	0.36	0.48	0.00	1.00
ANRR Democracy	8,623	0.44	0.50	0.00	1.00
CGV Democracy	8,900	0.45	0.50	0.00	1.00
BR Democracy	13,501	0.43	0.49	0.00	1.00
GWF Democracy	8,979	0.44	0.50	0.00	1.00
Polity Democracy	16,487	0.49	0.35	0.00	1.00
V-Dem Electoral Democracy	24,887	0.27	0.27	0.01	0.92
V-Dem Liberal Democracy	24,041	0.23	0.24	0.00	0.90
Freedom House Continuous Democracy	9,426	0.55	0.34	0.00	1.00
Freedom House Strict Democracy	9,426	0.40	0.49	0.00	1.00
Freedom House Flexible Democracy	9,426	0.71	0.45	0.00	1.00
Panel C: Accountability and Constraints					
V-Dem Vertical Accountability	7,596	0.61	0.25	0.00	1.00
V-Dem Horizontal Accountability	7,596	0.52	0.23	0.00	1.00
V-Dem Diagonal Accountability	7,596	0.62	0.23	0.00	1.00
Panel D: State Capacity					
H–S State Capacity	8,102	0.26	0.96	−2.31	2.96
WGI Government Effectiveness	4,524	−0.06	0.99	−2.44	2.47
V-Dem Fiscal Capacity	7,370	0.64	0.20	0.00	1.00
Panel E: Foreign Asset Positions					
EWN Log Total Foreign Assets per Capita	9,319	7.26	3.03	−0.12	18.64
EWN Log Outward FDI Stock per Capita	6,545	5.39	3.67	−7.31	17.48
EWN Log Portfolio Equity Stock per Capita	5,787	4.50	3.90	−11.02	17.99
Chinn–Ito Capital Account Openness	8,340	−0.00	1.53	−1.94	2.28
Panel F: Inequality					
SWIID Post-Fisc Gini	6,404	38.05	8.47	16.90	65.20
SWIID Pre-Fisc Gini	6,404	45.29	6.34	27.40	72.50
SWIID Absolute Redistribution	6,404	7.25	6.99	−5.50	25.80
SWIID Relative Redistribution	6,404	15.83	15.01	−13.55	52.26
Panel G: Economic and Demographic Characteristics					
WDI GDP per Capita Growth	11,223	1.95	6.49	−64.42	140.49
WDI Log GDP per Capita	11,345	8.41	1.51	4.81	12.42
WDI Oil Rents Share of GDP	8,347	3.79	9.85	0.00	87.09
WDI Trade Share of GDP	8,911	78.14	53.85	0.00	863.20
Barro–Lee Average Years of Schooling	1,898	5.45	3.18	0.02	13.18
WDI Log Population	14,094	14.83	2.44	7.91	21.10
WDI Urbanization	14,124	51.95	25.58	1.84	100.00

Notes: This table presents summary statistics for every variable used in the analysis, each summarized over all available observations in the form in which it enters. All controls enter as predetermined (pre-1995) values. Absolute redistribution is the pre-fisc minus the post-fisc Gini, and relative redistribution expresses that difference as a percent of the pre-fisc Gini.

A1.5 Variable Coding

This section lists the relevant categorical codings used in the analysis. Table A2 displays the list of English common-law country groups used as controls in Table 2. The coding of this variable is lifted from the legal-origin classification of La Porta, Lopez-de-Silanes, and Shleifer (2008). I obtain the data on the classification from the associated Quality of Government dataset. Table A3 lists the post-communist countries, entered as controls in Table 2. Table A4 lists the countries with no pre-1995 foreign citations, which receive no exposure share and thus do not contribute to identification. They are nonetheless included in the estimation sample. Table A5 lists the nine knowledge fields coded as information and communication technology and removed in the corresponding column of Table 2.

Table A2: Countries Coded as English Common-Law Legal Origin

AUS	Australia	GRD	Grenada	NPL	Nepal	TZA	Tanzania
BHS	Bahamas	GUY	Guyana	NZL	New Zealand	THA	Thailand
BHR	Bahrain	IND	India	NGA	Nigeria	TON	*Tonga*†
BGD	Bangladesh	IRL	Ireland	PAK	Pakistan	TTO	Trinidad and Tobago
BRB	Barbados	ISR	Israel	PNG	Papua New Guinea	TUV	Tuvalu
BTN	*Bhutan*†	JAM	Jamaica	WSM	Samoa	UGA	Uganda
BWA	Botswana	KEN	Kenya	SAU	Saudi Arabia	ARE	United Arab Emirates
CAN	Canada	KIR	*Kiribati*†	SLE	Sierra Leone	GBR	United Kingdom
CYP	Cyprus	LSO	*Lesotho*†	SGP	Singapore	USA	United States
DMA	Dominica	LBR	Liberia	SLB	*Solomon Islands*†	ZMB	Zambia
SWZ	Eswatini	MWI	Malawi	SOM	*Somalia*†	ZWE	Zimbabwe
FJI	*Fiji*†	MYS	Malaysia	ZAF	South Africa		
GMB	Gambia	MDV	Maldives	LKA	Sri Lanka		
GHA	Ghana	NRU	*Nauru*†	LCA	St. Lucia		

Notes: This table presents the 53 countries coded as having English common-law legal origin, used in column (3) of Table 2. The coding comes from La Porta, Lopez-de-Silanes, and Shleifer (2008)'s legal-origin classification. Legal origin is a time-invariant country characteristic, so the same list applies to every sample year. Of these countries, 45 appear in the estimation sample. The 8 shown in italics and marked † do not, almost always because they have no pre-1995 citation exposure shares.

Table A3: Countries Coded as Post-Communist

ALB	Albania	EST	Estonia	MDA	Moldova	SVK	Slovakia
ARM	Armenia	GEO	Georgia	MNG	Mongolia	SVN	Slovenia
AZE	Azerbaijan	HUN	Hungary	MNE	*Montenegro*†	TJK	Tajikistan
BLR	Belarus	KAZ	Kazakhstan	MKD	*North Macedonia*†	TKM	Turkmenistan
BIH	Bosnia and Herzegovina	XKX	*Kosovo*†	POL	Poland	UKR	Ukraine
BGR	Bulgaria	KGZ	Kyrgyzstan	ROU	Romania	UZB	Uzbekistan
HRV	Croatia	LVA	Latvia	RUS	Russia		
CZE	Czechia	LTU	Lithuania	SRB	*Serbia*†		

Notes: This table presents the 30 countries coded as post-communist, used in column (4) of Table 2. The list is hand-coded and includes the fifteen post-Soviet successor states, the Warsaw Pact members of Central and Eastern Europe, and the successor states of Yugoslavia, Albania, and Mongolia. The indicator is a time-invariant country characteristic, so the same list applies to every sample year. Of these countries, 26 are in the estimation sample. The 4 shown in italics and marked † do not due to a lack of pre-1995 citation exposure shares or the absence of a pre-1995 income control.

Table A4: Countries Excluded Due to Absence of Pre-1995 Foreign Citations

AGO	Angola	COM	Comoros	KIR	Kiribati	PLW	Palau
BEN	Benin	COD	Dem. Rep. Congo	LSO	Lesotho	RWA	Rwanda
BTN	Bhutan	DJI	Djibouti	FSM	Micronesia	SLB	Solomon Islands
KHM	Cambodia	GNQ	Equatorial Guinea	MNE	Montenegro	VCT	St. Vincent
CPV	Cape Verde	ERI	Eritrea	MOZ	Mozambique	TON	Tonga
CAF	Central African Rep.	FJI	Fiji	MKD	North Macedonia		

Notes: This table presents the 23 countries that appear in the estimation panel—having patent data and a democracy outcome—but record no pre-1995 foreign patent citations. Their exposure shares are therefore undefined, so they receive no weight in the shift-share estimator, as discussed in Section 4.2. The exclusion is a time-invariant country characteristic, so the same list applies to every sample year.

Table A5: Knowledge Fields Coded as ICT

Technology area	Four-digit IPC classes
Data and image recognition	B42D, G06K, G06V
Imaging, video, and broadcast	G03B, G06T, G11B, H04H, H04N
Analog control and computing	G05D, G06D
Mechanical computing	G06C
Counting devices	G06M
Digital communication and networks	G09C, H04K, H04L, H04Q, H04W
Oscillators and pulse circuits	H03B, H03K, H03L
Signal-processing and transmission circuits	H03C, H03D, H03F, H03G, H03H, H03J, H04B, H04J
Audio transducers	H04R, H04S

Notes: This table presents the nine knowledge fields whose four-digit IPC members are in majority drawn from the information- and communication-technology classes: G06 (computing), G11 (information storage), H03 (basic electronic circuitry), H04 (electric communication), and H01L (semiconductors). These are the fields removed from the instrument in the *Excluding ICT fields* column of Table 2. The technology-area column is a descriptive label, of my own construction, for each field's dominant classes.

A2 The Identifying Variation of the Instrument

This section details several features of the identifying variation I exploit with my instrument, which rests on the exogeneity of global patenting growth with respect to country-level conditions.

Table A6 reports descriptive statistics for the shifts. In Panel (A), I group the shifts by their broad IPC technology area and report each area's number of fields, share of the total exposure weight, and mean and standard deviation of its shifts. Panel (B) displays the shifts' distribution across field-years, as well as the distribution weighted by exposure and residualized on year fixed effects. Panel (C) describes the concentration of the exposure weights, reporting the effective number of independent shifts across both the fields and the eight IPC technology areas.

Table A7 reports the shock-level balance tests. Following BHJ, I regress each predetermined country characteristic on the instrument (after conditioning on pre-1995 income and initial democracy, as in my preferred specification). Most variables are balanced, the exceptions being correlates of development—log GDP per capita, population, and schooling. As might be expected, more exposed countries have modestly higher levels of development than less exposed countries. The controls in Table 1 condition on these variables, and the results are qualitatively unchanged. State capacity and foreign asset positions additionally show a pre-sample trend correlated with the instrument.

Table A8 probes these pre-trends. It confirms that exposure to faster-growing fields is correlated with the pre-sample (1985–1994) trends in state capacity and foreign asset positions. I then re-estimate each outcome with the two pre-trends added as predetermined controls, at the preferred specification of Table 1. The democracy null is insignificant throughout, and the capacity–accountability gap remains roughly unchanged in magnitude. The WGI government effectiveness measure remains positive and significant. The effect on foreign asset positions attenuates but keeps its sign. Only the Hanson–Sigman capacity *level* falls to near zero.

Table A6: Shift Descriptive Statistics

Panel A. Shifts by IPC technology area				
	Fields	Weight (%)	Mean	SD
A. Human necessities	40	16.7	7.35	1.53
B. Performing operations; transport	80	21.9	6.91	1.83
C. Chemistry; metallurgy	22	19.4	7.09	2.28
D. Textiles; paper	20	3.2	5.67	1.77
E. Fixed constructions	17	8.4	7.59	1.21
F. Mechanical engineering	40	10.2	6.88	1.87
G. Physics	27	11.3	7.00	2.55
H. Electricity	16	8.9	8.33	1.98
All fields	262	100.0	7.03	1.97

Panel B. Distribution of the shifts and the instrument			
	Observations	Mean	SD
Shifts	5,517	7.03	1.97
Weighted by exposure share	5,500	8.87	1.46
Residualized on year fixed effects	5,517	0.00	1.93
Instrument	3,780	8.87	0.78

Panel C. Concentration of exposure weights	
	Value
Knowledge fields	262
Effective number of fields	66
Effective number of technology areas	6.6

Notes: This table presents summary statistics for the shifts that drive the instrument. A shift is the log of worldwide patenting in a knowledge field in a given year, and a field's exposure weight is its mean share across all countries. The weighted row omits the 17 field-years whose field carries no weight. The effective number of independent shifts is the inverse Herfindahl index of the exposure weights.

Table A7: Shock Balance Tests

Balance variable	Coef.	SE	Countries
Log GDP per capita	$+0.035^{*}$	(0.019)	178
Trade / GDP	$+0.022$	(0.014)	146
Capital-account openness	-0.009	(0.015)	143
Average years of schooling	$+0.037^{**}$	(0.018)	139
Log population	$+0.052^{***}$	(0.014)	178
State capacity	$+0.001$	(0.010)	154
Democracy	-0.017	(0.012)	178
Oil rents / GDP	-0.005	(0.013)	164
Income inequality	-0.040	(0.025)	126
Foreign assets per capita	$+0.001$	(0.006)	163
Democracy pre-trend, 1985–1994	$+0.016$	(0.038)	132
State capacity pre-trend, 1985–1994	$+0.067^{***}$	(0.025)	133
Foreign assets pre-trend, 1985–1994	$+0.071^{**}$	(0.031)	141
Patenting per capita pre-trend, 1985–1994	$+0.062^{**}$	(0.030)	140
GDP per capita growth, 1985–1994	$+0.004$	(0.033)	160

Notes: This table presents shock-level balance tests for the instrument, following Borusyak, Hull, and Jaravel (2022). Each row regresses a predetermined, standardized country characteristic on the instrument, conditioning on pre-1995 income and initial democracy. Pre-trend rows use the change in a variable's mean between the halves of the stated window. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A8: Pre-Period Trends

Panel A. The instrument and pre-period trends, by window				
	1970–1979	1980–1989	1985–1994	1970–1994
Democracy	−0.001	−0.048*	0.016	−0.022
	(0.014)	(0.026)	(0.037)	(0.016)
State capacity	−0.047***	0.016	0.064***	0.020
	(0.014)	(0.017)	(0.022)	(0.015)
Foreign asset positions	−0.003	0.008	0.067**	0.001
	(0.016)	(0.016)	(0.030)	(0.018)

Panel B. Estimates controlling for the pre-period trend				
	Preferred	Same sample	+ 1985–94 trend	+ 1970–94 trend
Democracy	−0.029	−0.041	−0.060	−0.014
	(0.045)	(0.068)	(0.058)	(0.056)
First-stage F	23.8	15.5	15.6	17.7
Observations	2,997	2,240	2,240	2,240
State capacity	0.158***	0.170**	0.028	0.150*
	(0.060)	(0.079)	(0.092)	(0.079)
First-stage F	28.5	28.6	21.2	31.1
Observations	2,668	2,261	2,261	2,261
Foreign asset positions	0.081***	0.120***	0.040	0.120***
	(0.031)	(0.032)	(0.035)	(0.031)
First-stage F	28.7	21.2	9.6	30.0
Observations	2,909	2,253	2,253	2,253
Capacity–accountability gap	0.225***	0.191**	0.180*	0.193*
	(0.069)	(0.093)	(0.106)	(0.103)
First-stage F	21.2	18.5	16.0	17.4
Observations	2,625	2,239	2,239	2,239

Notes: This table presents pre-trend tests for the instrument. A pre-period trend is the change in a country's mean value between the halves of the stated window, measured on Polity for democracy, which is graded where the outcome is binary, and on state capacity for the gap. Panel A regresses that trend on the instrument with the controls of Table 1, column (3). Panel B adds the trend to that specification, where the *same sample* column restricts to countries with both windows available but adds no control. The capacity–accountability gap is Hanson–Sigman capacity minus V-Dem vertical accountability, both standardized. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3 Robustness

A3.1 Alternative Measures of Innovation

Because the baseline treatment is a patent count, I re-estimate the main specification with four quality-adjusted alternatives from Trajtenberg, Henderson, and Jaffe (1997)—citation-weighted, originality-weighted, generality-weighted, and family-size-weighted counts—each expressed per capita and logged. Table A9 shows these correlate between 0.95 and 0.98 with the raw count. Table A10 shows the estimates barely move, with all results mirroring those of the main text. The instrument is held fixed at the raw count-based shift-share throughout, since citation-weighted shifts would embed post-1995 information and no longer be predetermined.

Table A9: Pairwise Correlations Among Innovation Measures

	Patent counts	Citation-weighted	Originality-weighted	Generality-weighted	Family-size-weighted
Patent counts	1.00				
Citation-weighted	0.95	1.00			
Originality-weighted	0.95	0.95	1.00		
Generality-weighted	0.96	0.97	0.98	1.00	
Family-size-weighted	0.98	0.97	0.96	0.96	1.00

Notes: This table presents pairwise correlations among the innovation measures, computed with pairwise deletion. Each is logged and expressed per million residents, with zeros dropped, and the weights follow Trajtenberg, Henderson, and Jaffe (1997).

Table A10: Alternative Measures of Innovation

	Measure of innovation				
	Patent counts (1)	Citation-weighted (2)	Originality-weighted (3)	Generality-weighted (4)	Family-size-weighted (5)
Panel A. Democracy					
Log patents per million	−0.029	−0.029	−0.037	−0.057	−0.028
	(0.045)	(0.047)	(0.058)	(0.054)	(0.045)
Mean of outcome	0.627	0.624	0.646	0.650	0.624
Observations	2,997	3,067	2,723	2,684	3,067
Countries	183	183	179	180	183
First-stage F	23.8	20.9	13.5	15.2	25.6
Panel B. State capacity					
Log patents per million	0.158***	0.159***	0.188**	0.148**	0.157***
	(0.060)	(0.055)	(0.076)	(0.067)	(0.056)
Mean of outcome	0.006	0.005	0.007	0.007	0.005
Observations	2,668	2,726	2,449	2,411	2,726
Countries	155	155	153	153	155
First-stage F	28.5	35.2	25.8	29.6	32.2
Panel C. Foreign asset positions					
Log patents per million	0.081***	0.079***	0.100***	0.087***	0.080***
	(0.031)	(0.029)	(0.033)	(0.033)	(0.029)
Mean of outcome	−0.120	−0.118	−0.121	−0.123	−0.118
Observations	2,909	2,973	2,642	2,602	2,973
Countries	176	176	173	174	176
First-stage F	28.7	26.6	16.7	20.3	30.2

Notes: This table presents IV estimates of innovation's effect on democracy, state capacity, and capital mobility across alternative measures of the treatment. Each column estimates the preferred specification of Table 1, column (3), with the given patent variant, logged and per million residents, following Trajtenberg, Henderson, and Jaffe (1997). All columns use the same count-based $k = 263$ shift-share as the instrument. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.2 Alternative Measures of Democracy

This section investigates alternative measures of democracy. I use measures from the following sources: Boix, Miller, and Rosato (2013) [BMR], Acemoglu et al. (2019) [ANRR], Cheibub, Gandhi, and Vreeland (2010) [CGV], Bjørnskov and Rode (2020) [BR], Geddes, Wright, and Frantz (2014) [GWF], Freedom House [FH], the Polity5 project [Polity], and the Varieties of Democracy project [V-Dem]. Following Blattman, Gehlbach, and Yu (2025, p. 11), I lead the GWF measure by one period. I also construct two dichotomous indices using existing Freedom House data. The first, strict coding, categorizes countries as democratic only if they are marked as "free", while the second, flexible coding also counts "partly free" countries as democratic.

Table **??** shows the correlation matrix of the measures, and they are all highly correlated with one another. Table A12 re-estimates my preferred specification with each measure as the outcome. All results come back null.

Table A11Pairwise correlations among democracy measures

	ANRR	BMR	BR	CGV	GWF	FH continuous	FH flexible	FH strict
ANRR	1.000							
BMR	0.898	1.000						
BR	0.889	0.894	1.000					
CGV	0.882	0.907	0.969	1.000				
GWF	0.882	0.892	0.906	0.923	1.000			
FH continuous	0.847	0.853	0.822	0.826	0.834	1.000		
FH flexible	0.664	0.641	0.638	0.609	0.615	0.800	1.000	
FH strict	0.750	0.767	0.723	0.743	0.751	0.860	0.529	1.000
Polity	0.904	0.828	0.842	0.844	0.863	0.884	0.693	0.736
V-Dem electoral	0.854	0.848	0.829	0.833	0.837	0.923	0.671	0.824
V-Dem liberal	0.819	0.830	0.805	0.807	0.821	0.919	0.640	0.846

Notes: Correlations are computed after pairwise deletion of missing observations.

Table A12: The Democracy Null Across Alternative Democracy Measures

	Democracy measure used as the outcome										
	Dichotomous							Continuous			
	BMR	ANRR	CGV	BR	GWF	FH_s	FH_f	Polity	V-Dem_e	V-Dem_l	FH_c
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Log patents per million	−0.029	0.018	−0.069	−0.017	−0.058	−0.013	0.030	−0.033	−0.015	−0.010	0.002
	(0.045)	(0.039)	(0.051)	(0.048)	(0.063)	(0.032)	(0.038)	(0.034)	(0.024)	(0.020)	(0.025)
Mean of outcome	0.627	0.660	0.636	0.639	0.631	0.478	0.771	0.702	0.553	0.444	0.629
Observations	2,997	2,185	1,935	2,947	1,756	2,988	2,988	2,656	2,789	2,771	2,988
Countries	183	176	180	180	142	182	182	156	164	163	182
First-stage F	23.8	24.1	18.0	20.3	14.7	23.6	24.3	13.5	25.1	27.0	27.8

Notes: This table presents IV estimates of innovation's effect on democracy across alternative measures of the outcome. Each column estimates the preferred specification of Table 1, column (3), with the given measure in its natural scale. Column (1) is the estimate in Table 1. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.3 Alternative Measures of State Capacity

Table A14 re-estimates the capacity result against five alternatives to the Hanson–Sigman index: the World Bank government effectiveness index, V-Dem's fiscal capacity and administration variable, tax revenue, and territorial control. All measures are standardized. Table A13 shows the two composite indices are strongly correlated, while the component indices are less associated. The effects of innovation remain similar to those in the main text, with both composite indices recovering positive coefficients, significant at the one percent level. The components are consistently signed but imprecise, as expected if innovation raises general administrative capacity rather than any single part of that capacity.

Table A13: Pairwise Correlations Among State-Capacity Measures

	H–S	WGI	Fiscal	Admin.	Tax/ GDP	Territory
H–S	1.00					
WGI	0.90	1.00				
Fiscal	0.77	0.72	1.00			
Admin.	0.83	0.81	0.64	1.00		
Tax/GDP	0.67	0.54	0.49	0.55	1.00	
Territory	0.59	0.49	0.41	0.43	0.36	1.00

Notes: This table presents pairwise correlations among the state-capacity measures, computed with pairwise deletion. The measures are the Hanson–Sigman index (H–S), WGI government effectiveness, V-Dem fiscal capacity (Fiscal) and impartial public administration (Admin.), tax revenue over GDP, and territorial control.

Table A14: Innovation and State Capacity Across Alternative Measures

	State-capacity measure used as the outcome					
	Composite indices		Components			
	H–S (1)	WGI (2)	Fiscal (3)	Admin. (4)	Tax/GDP (5)	Territory (6)
Log patents per million	0.158***	0.124***	0.071	0.055	0.052	0.163*
	(0.060)	(0.030)	(0.068)	(0.048)	(0.074)	(0.099)
Mean of outcome	0.006	0.010	0.014	0.005	−0.003	−0.007
Observations	2,668	2,344	2,661	2,801	2,760	2,801
Countries	155	168	156	164	165	164
First-stage F	28.5	34.5	10.1	35.3	27.4	28.6

Notes: This table presents IV estimates of innovation's effect on state capacity across alternative measures of the outcome. Each column estimates the preferred specification of Table 1, column (3), with the given measure standardized. Column (1) is the estimate in Table 1. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.4 Alternative Measures of Capital Mobility

Table A16 reports seven measures of capital mobility, since the mechanism concerns capital that can credibly leave rather than "financial openness" in general: the headline foreign-asset stock, the same net of official reserves, the position over GDP, three private asset classes, and the *de jure* Chinn–Ito index. Table A15 shows the *de facto* positions move together closely while the *de jure* index is an outlier. The estimates support the result in sign and rough magnitude but qualify it: netting out central-bank reserves costs two-fifths of the point estimate and its significance (0.081 to 0.048), and the surviving private evidence is concentrated in the most liquid claim, deposits and loans (0.066, significant), rather than spread across the balance sheet. The *de jure* index gives 0.086 but far less precisely—consistent with innovation making capital more mobile in fact rather than inducing governments to liberalize the capital account.

Table A15: Pairwise Correlations Among Capital-Mobility Measures

	Assets per capita	Private assets per capita	Assets as share of GDP	Direct investment	Portfolio debt	Deposits and loans	Chinn–Ito
Assets per capita	1.00						
Private assets per capita	0.99	1.00					
Assets as share of GDP	0.92	0.89	1.00				
Direct investment	0.91	0.91	0.79	1.00			
Portfolio debt	0.88	0.88	0.78	0.82	1.00		
Deposits and loans	0.98	0.98	0.89	0.89	0.85	1.00	
Chinn–Ito	0.57	0.57	0.50	0.55	0.49	0.53	1.00

Notes: This table presents pairwise correlations among the capital-mobility measures, computed with pairwise deletion. The External Wealth of Nations positions are in logs, per capita except where scaled by GDP, and the Chinn–Ito index is in levels. Private assets are foreign assets net of official reserves.

Table A16: Innovation and Capital Mobility Across Alternative Measures

	Financial-integration measure used as the outcome						
	Aggregate positions			By asset class			*De jure*
	Assets per capita	Private assets per capita	Assets as share of GDP	Direct investment	Portfolio debt	Deposits and loans	Chinn–Ito
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log patents per million	0.081***	0.048	0.085*	0.086**	−0.007	0.066**	0.086
	(0.031)	(0.030)	(0.050)	(0.037)	(0.032)	(0.028)	(0.084)
Mean of outcome	−0.120	−0.036	−0.133	−0.043	0.022	−0.132	−0.006
Observations	2,909	2,888	2,909	2,323	1,913	2,892	2,808
Countries	176	175	176	132	105	175	173
First-stage F	28.7	25.9	28.4	18.3	16.3	27.8	27.2

Notes: This table presents IV estimates of innovation's effect on capital mobility across alternative measures of the outcome. Each column estimates the preferred specification of Table 1, column (3), with the given measure standardized. Columns (1)–(6) are External Wealth of Nations foreign-asset stocks and column (7) is the *de jure* Chinn–Ito index. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.5 Alternative Inference

Table A17 and Table A18 re-run Tables 1 and 2 with Anderson–Rubin inference, which produces valid confidence intervals even if the instrument is weak. The results are very similar to the baselines.

Table A17: Anderson–Rubin Inference for the Main Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Panel A. Democracy				
Log patents per million	−0.016	−0.065	−0.029	−0.015	−0.014	0.015	0.046	0.048
AR 95% CI	[−.21,.05]	[−.23,.04]	[−.14,.06]	[−.17,.10]	[−.15,.11]	[−.18,.17]	[−.30,.32]	[−.43,.30]
AR p-value	0.71	0.22	0.51	0.80	0.82	0.86	0.74	0.74
Mean of outcome	0.626	0.629	0.627	0.629	0.627	0.664	0.664	0.664
Observations	3,055	3,007	2,997	2,609	2,601	2,291	2,291	2,291
Countries	187	184	183	151	149	126	126	126
First-stage F	10.8	18.2	23.8	18.9	29.2	27.3	13.4	9.2
				Panel B. State capacity				
Log patents per million	0.229**	0.170**	0.158**	0.157**	0.158**	0.166*	0.206	0.165
AR 95% CI	[.07,.30]	[.03,.28]	[.03,.29]	[.01,.31]	[.01,.30]	[−.01,.37]	[−.05,.57]	[−.11,.50]
AR p-value	0.02	0.02	0.02	0.04	0.03	0.07	0.11	0.23
Mean of outcome	0.000	0.006	0.006	0.054	0.054	0.144	0.144	0.144
Observations	2,715	2,688	2,668	2,489	2,489	2,227	2,227	2,227
Countries	159	157	155	141	141	122	122	122
First-stage F	15.4	21.5	28.5	29.3	48.3	48.9	23.4	20.3
				Panel C. Foreign asset positions				
Log patents per million	0.161	0.026	0.081**	0.069*	0.069*	0.092*	0.086	0.069
AR 95% CI	[−.16,.27]	[−.10,.12]	[.01,.14]	[−.01,.14]	[−.02,.13]	[−.01,.18]	[−.12,.23]	[−.18,.24]
AR p-value	0.17	0.62	0.03	0.08	0.10	0.07	0.30	0.46
Mean of outcome	0.000	-0.071	-0.120	-0.158	-0.158	-0.129	-0.129	-0.129
Observations	3,175	3,049	2,910	2,576	2,575	2,267	2,267	2,267
Countries	199	190	177	149	148	125	125	125
First-stage F	10.8	23.1	28.8	22.5	28.7	29.8	13.0	10.6
Log GDP per capita		✓	✓	✓	✓	✓	✓	✓
Initial outcome level			✓	✓	✓	✓	✓	✓
Initial democracy			✓	✓	✓	✓	✓	✓
Trade and *de jure* capital openness				✓	✓	✓	✓	✓
Oil rents					✓	✓	✓	✓
Years of schooling						✓	✓	✓
Log population							✓	✓
Region fixed effects								✓
Year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Effective number of shifts	99	99	99	99	99	99	99	99

Notes: This table presents IV estimates of innovation's political and economic effects under Anderson–Rubin inference, reproducing Table 1. Anderson–Rubin 95 percent confidence sets, obtained by grid-inverting the field-clustered AR test, are in brackets with the p-value beneath, and stars follow the AR test. They remain valid regardless of instrument strength. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A18: Anderson–Rubin Inference for the Robustness Battery

	Unit-level covariates								Shock-side checks	
	Baseline (1)	Initial outcome × year (2)	Common-law dummies × year (3)	Post-communist × year (4)	Region × year (5)	Manufacturing share (6)	Inward FDI stock (7)	Spatial lags (8)	Section × year (9)	Excluding ICT fields (10)
Panel A. Democracy										
Log patents per million	−0.029	−0.018	−0.029	−0.048	−0.055	−0.077	−0.023	−0.060	−0.036	−0.026
AR 95% CI	[−.14,.06]	[−.16,.08]	[−.14,.06]	[−.20,.06]	[−.26,.07]	[−.31,.09]	[−.13,.07]	[−.18,.03]	[−.31,.10]	[−.15,.07]
AR p-value	0.51	0.73	0.51	0.39	0.41	0.36	0.61	0.19	0.61	0.60
Mean of outcome	0.627	0.624	0.625	0.625	0.625	0.663	0.626	0.633	0.627	0.627
Observations	2,997	2,955	2,961	2,961	2,961	2,664	2,869	2,823	2,997	2,997
Countries	183	178	180	180	180	157	173	159	183	183
First-stage F	23.8	19.6	25.1	19.5	16.7	14.6	25.3	22.0	9.8	22.7
Panel B. State capacity										
Log patents per million	0.158**	0.176**	0.153**	0.177**	0.164*	0.189**	0.174***	0.158*	0.231**	0.168**
AR 95% CI	[.03,.29]	[.01,.41]	[.03,.28]	[.03,.30]	[−.02,.32]	[.01,.41]	[.06,.29]	[−.01,.30]	[.05,.39]	[.03,.31]
AR p-value	0.02	0.04	0.02	0.03	0.08	0.03	0.01	0.06	0.02	0.02
Mean of outcome	0.006	0.011	0.011	0.011	0.011	0.079	0.022	0.051	0.006	0.006
Observations	2,668	2,661	2,661	2,661	2,661	2,382	2,625	2,543	2,668	2,668
Countries	155	154	154	154	154	134	152	138	155	155
First-stage F	28.5	14.1	33.8	35.5	28.7	22.8	34.9	28.9	16.2	25.8
Panel C. Foreign asset positions										
Log patents per million	0.081**	0.091**	0.084**	0.113**	0.100*	0.087*	0.071*	0.112**	0.113**	0.078*
AR 95% CI	[.01,.14]	[.00,.16]	[.02,.13]	[.02,.19]	[−.01,.20]	[−.01,.20]	[−.01,.13]	[.00,.20]	[.01,.22]	[−.01,.15]
AR p-value	0.03	0.04	0.02	0.02	0.08	0.08	0.08	0.05	0.04	0.06
Mean of outcome	-0.120	-0.128	-0.128	-0.128	-0.128	-0.089	-0.139	-0.099	-0.120	-0.120
Observations	2,909	2,873	2,873	2,873	2,873	2,610	2,864	2,735	2,909	2,909
Countries	176	174	174	174	174	152	173	154	176	176
First-stage F	28.7	18.7	31.1	24.9	19.9	22.0	25.2	23.0	11.7	28.7

Notes: This table presents IV estimates of the robustness battery under Anderson–Rubin inference, reproducing Table 2. Anderson–Rubin 95 percent confidence sets, obtained by grid-inverting the field-clustered AR test, are in brackets with the p-value beneath, and stars follow the AR test. They remain valid regardless of instrument strength. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.6 Identification and the Instrument

In this section, I implement three checks to probe the source of the instrument's identifying variation. Table A19 replaces the worldwide field shift I use in the main text with a leave-one-out variant that removes each country's own patents from the field total before forming its instrument. The estimates are essentially unchanged, suggesting that the results are not driven by countries' own patents. Table A20 implements a similar check, instead dropping each IPC technology section from the instrument in turn to see if a particular type of technology drives the findings. The results indicate that this is not the case, remaining similar to my baseline estimates. Finally, because the leave-one-out instrument has no shock-level representation, Table A21 rebuilds the shifts from worldwide patenting that excludes the three largest producers (China, Japan, and the United States) as a shock-level alternative; the estimates are again essentially unchanged.

Table A19: Leave-One-Out Robustness

	Democracy	State capacity	Foreign asset positions
	(1)	(2)	(3)
Worldwide shift	−0.030	0.160**	0.081**
	(0.063)	(0.064)	(0.035)
Leave-out shift	−0.039	0.182**	0.082**
	(0.075)	(0.073)	(0.041)
First-stage F, worldwide	12.7	12.9	13.6
First-stage F, leave-out	10.7	11.7	12.0
Countries	162	139	156
Observations	2,859	2,550	2,771

Notes: This table presents IV estimates of innovation's effects at the preferred specification of Table 1, column (3), comparing the worldwide field shift with a leave-own-out version that removes each country's own patents from the field total. Standard errors, clustered at the country level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A20: Leave-One-Section-Out Robustness

Excluded section:	None (1)	Human necessities (2)	Transport (3)	Chemistry (4)	Textiles (5)	Construction (6)	Mechanical (7)	Physics (8)	Electricity (9)
Panel A. Democracy									
Log patents per million	−0.029	−0.027	−0.020	−0.069	−0.036	−0.029	−0.013	0.004	−0.040
	(0.045)	(0.043)	(0.050)	(0.052)	(0.043)	(0.044)	(0.051)	(0.050)	(0.052)
First-stage F	23.8	23.9	19.3	15.1	30.9	24.0	16.0	18.6	17.8
Panel B. State capacity									
Log patents per million	0.158***	0.165***	0.179***	0.087	0.145***	0.159***	0.146*	0.189***	0.190***
	(0.060)	(0.062)	(0.065)	(0.065)	(0.052)	(0.060)	(0.079)	(0.070)	(0.062)
First-stage F	28.5	26.3	24.5	16.7	40.6	29.1	18.5	27.3	20.8
Panel C. Foreign asset positions									
Log patents per million	0.081***	0.077**	0.065*	0.114***	0.074**	0.083***	0.084**	0.089**	0.072*
	(0.031)	(0.031)	(0.033)	(0.027)	(0.030)	(0.031)	(0.033)	(0.041)	(0.038)
First-stage F	28.7	25.5	24.2	19.3	38.3	27.9	20.3	23.4	21.7

Notes: This table presents IV estimates of innovation's effects at the preferred specification of Table 1, column (3), with each column dropping one broad IPC technology section from the instrument. Column (1) uses all fields. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A21: Split-Sample Instrument Construction

	Democracy	State capacity	Foreign asset positions
	(1)	(2)	(3)
Worldwide shifts (baseline)	−0.029	0.158***	0.081***
	(0.045)	(0.060)	(0.031)
Shifts excluding China, Japan, United States	−0.035	0.159**	0.072**
	(0.052)	(0.064)	(0.036)

Notes: This table presents 2SLS estimates at the preferred specification (Table 1, column 3) when the instrument's shifts are rebuilt from worldwide patenting that excludes the three largest producers (China, Japan, and the United States), for all countries. Because the leave-one-out instrument has no shock-level representation, this split-sample construction provides a shock-level alternative that severs the mechanical link between a country's own patenting and the shifts. Estimation is at the shock level with exposure-robust standard errors clustered by knowledge field. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.7 The Exclusion Restriction

In this section, I implement two checks that test the plausibility of the exclusion restriction. Table A22 splits the knowledge fields into disjoint halves (even fields in one half and odd in the other) and estimates the effects of interest using each half as an instrument. A Hansen overidentification test does not reject that the two halves identify the same parameter, supporting the exclusion restriction.

Table A23 follows Conley, Hansen, and Rossi (2012), granting the instrument its own direct effect. The democracy null and the capacity–democracy gap are robust across any plausible leakage, but the individual capacity and foreign-asset estimates lose significance once the assumed leakage reaches roughly one-fifth of the instrument's reduced-form effect on growth.

Table A22: Field-Split Overidentification Test

	Odd half	Even half	Both	p (diff.)	p (diff., ER)	Hansen J	p (Hansen)	N
Democracy	−0.044	−0.177	−0.110	0.36	0.33	0.92	0.40	2,764
	(0.088)	(0.114)	(0.081)					
State capacity	0.140**	−0.017	0.106*	0.22	0.20	1.43	0.24	2,513
	(0.063)	(0.109)	(0.057)					
Foreign asset positions	0.073	0.079	0.076*	0.92	0.91	0.01	0.99	2,688
	(0.048)	(0.049)	(0.040)					

Notes: This table presents IV estimates of innovation's effects from two disjoint halves of the instrument, built from the odd and even knowledge fields and compared by a difference test and the Hansen J. Estimation is at the country level, since two instruments have no shock-level form. Standard errors, clustered at the country level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A23: Plausibly Exogenous Bounds

	Assumed direct effect γ/γ^*				
Outcome	0.0	0.5	1.0	2.0	Breakdown γ/γ^*
Democracy	-0.080 [-0.27, $+0.11$]	-0.158 [-0.36, $+0.05$]	-0.237 [-0.46, -0.01]	-0.394 [-0.66, -0.13]	none
State capacity	$+0.155$ [$+0.04$, $+0.27$]	$+0.076$ [-0.05, $+0.20$]	-0.004 [-0.14, $+0.13$]	-0.162 [-0.33, $+0.00$]	0.2
Foreign asset positions	$+0.089$ [$+0.03$, $+0.15$]	$+0.015$ [-0.06, $+0.09$]	-0.058 [-0.15, $+0.03$]	-0.205 [-0.35, -0.06]	0.2
Capacity–democracy gap	$+0.261$ [$+0.09$, $+0.43$]	$+0.186$ [$+0.02$, $+0.35$]	$+0.111$ [-0.05, $+0.27$]	-0.039 [-0.21, $+0.13$]	0.7

Notes: This table presents IV estimates of innovation's effects when the instrument is granted a direct effect γ on the outcome, following Conley, Hansen, and Rossi (2012). Leakage is expressed relative to the benchmark $\gamma^* = 0.023$, the instrument's reduced-form effect on growth. The breakdown column reports the smallest γ/γ^* at which the estimate loses significance. Exposure-robust 95 percent confidence intervals, clustered at the field level, are in brackets.

A3.8 Robustness to Influential Observations

Table A24 tests whether my results are determined by influential outlier countries or fields. I drop the largest patent producing countries (Panel A), the smallest fields and outlier shift-years (Panel B), and thin citation-base countries (Panel C). In all cases, the democracy null and positive effect on mobility hold up. State capacity is weakened but never changes sign.

Table A24: Robustness to Influential Observations and Instrument Construction

	Democracy	State capacity	Foreign asset positions
	(1)	(2)	(3)
Panel A. Influential countries			
Baseline	−0.030	0.160***	0.081***
	(0.046)	(0.062)	(0.031)
Excluding China	−0.029	0.166**	0.081**
	(0.049)	(0.067)	(0.033)
Excluding China, Japan, United States	−0.032	0.174***	0.083**
	(0.051)	(0.067)	(0.035)
Excluding the five largest producers	−0.054	0.186**	0.083**
	(0.061)	(0.076)	(0.041)
Panel B. Small-field tail			
Drop smallest 10% of fields	−0.017	0.168***	0.082***
	(0.044)	(0.062)	(0.031)
Drop smallest 25% of fields	−0.025	0.073	0.062**
	(0.037)	(0.054)	(0.031)
Drop outlier shift-years ($\tilde{g} < -2$)	−0.007	0.169***	0.083***
	(0.043)	(0.063)	(0.031)
Panel C. Thin pre-1995 citation bases			
Drop $<$ 5 nonzero fields	−0.070	0.087	0.066**
	(0.047)	(0.062)	(0.028)
Drop $<$ 10 nonzero fields	−0.075**	0.023	0.072***
	(0.036)	(0.040)	(0.023)
Drop max share $>$ 0.5	−0.101**	0.074*	0.076***
	(0.045)	(0.044)	(0.027)

Notes: This table presents IV estimates of innovation's effects at the preferred specification of Table 1, column (3), re-estimated on trimmed samples. Each row drops the countries carrying most of the exposure-share mass (Panel A), the smallest fields or outlier shift-years (Panel B), or countries with thin pre-1995 citation bases (Panel C). Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.9 Additional Outcomes: Inequality

Table A25 tests the inequality mechanism implied by my theory, re-estimating the preferred specification with four measures of income inequality from the Standardized World Income Inequality Database. These include net (post-fisc) Gini, market (pre-fisc) Gini, and two measures of redistribution. Innovation does not show a significant association with any of the measures.

Table A25: Innovation and Income Inequality

	Net (disposable) Gini (1)	Market (gross) Gini (2)	Absolute redistribution (3)	Relative redistribution (4)
Log patents per million	0.026	0.031	−0.040	−0.054
	(0.052)	(0.058)	(0.048)	(0.051)
Mean of outcome	-0.002	0.023	0.021	0.019
Observations	2,493	2,493	2,493	2,493
Countries	146	146	146	146
First-stage F	15.9	18.9	16.4	15.1
Log GDP per capita	✓	✓	✓	✓
Initial outcome level	✓	✓	✓	✓
Initial democracy	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓

Notes: This table presents IV estimates of innovation's effect on income inequality. Each column estimates the preferred specification of Table 1, column (3), with the given SWIID measure standardized. Exposure-robust standard errors, clustered at the field level, are in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A3.10 Heterogeneity in the Democracy Effect

This section investigates whether the pooled null hides offsetting effects across countries. Table A26 splits the sample at pre-1995 terciles of six moderators: initial regime type, income, inequality, state capacity, capital openness, and resource dependence. Because these interactions inherently reduce power, Anderson–Rubin sets in Panel B should be treated as primary. No subgroup returns a positive effect on democracy. Sixteen of the eighteen confidence sets contain zero, and the two that exclude zero fall fully below zero. The exception is the anocracy cell, where the instrument is too weak to be informative.

Table A26: Heterogeneity in the Democracy Effect

	Regime type	Income	Inequality (net Gini)	State capacity	Capital openness	Resource dependence
Panel A. Conventional IV (coefficient, clustered SE)						
Bottom tercile	−0.118 (0.123)	−0.065 (0.129)	0.115 (0.079)	−0.204 (0.190)	−0.075 (0.181)	−0.112 (0.159)
Middle tercile	−0.418 (0.406)	−0.264 (0.207)	−0.200* (0.106)	0.029 (0.118)	−0.147 (0.130)	−0.095 (0.114)
Top tercile	−0.056 (0.038)	−0.041 (0.074)	−0.026 (0.382)	0.018 (0.028)	−0.086 (0.111)	−0.991 (1.409)
Panel B. Anderson–Rubin 95% confidence set						
Bottom tercile	[−0.50, 0.13]	[−0.75, 0.39]	[−0.27, 0.28]	[−1.26, 0.13]	[−3.64, 0.18]	[−2.01, 0.08]
Middle tercile	$(-\infty, \infty)$	$(-\infty$, −0.04]	[−0.66, −0.04]	[−0.18, 0.38]	[−1.07, 0.06]	[−0.30, 0.64]
Top tercile	[−0.15, 0.02]	[−0.34, 0.07]	$(-\infty, \infty)$	[−0.05, 0.07]	[−1.00, 0.06]	$(-\infty, \infty)$
First-stage F						
Bottom tercile	9.6	4.8	6.1	6.3	4.5	5.2
Middle tercile	0.9	3.5	9.7	14.4	5.8	5.8
Top tercile	16.4	9.9	0.9	22.6	6.0	0.5

Notes: This table presents IV estimates of innovation's effect on democracy within predetermined subgroups. Each cell estimates the preferred specification of Table 1, column (3) on the given pre-1995 tercile of the moderator, with the Boix–Miller–Rosato indicator as the outcome. For regime type the terciles are autocracies, anocracies, and democracies by pre-1995 Polity. Education splits are omitted, as the instrument is uninformative in all three terciles ($F < 2$). Panel A reports exposure-robust standard errors, clustered at the field level, in parentheses. Panel B reports Anderson–Rubin 95 percent confidence sets, where $(-\infty, \infty)$ marks an uninformative instrument. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.